



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING

SEMESTER 2

2023/2024

SECP2613 - SYSTEM ANALYSIS AND DESIGN (WBL)

SECTION 01

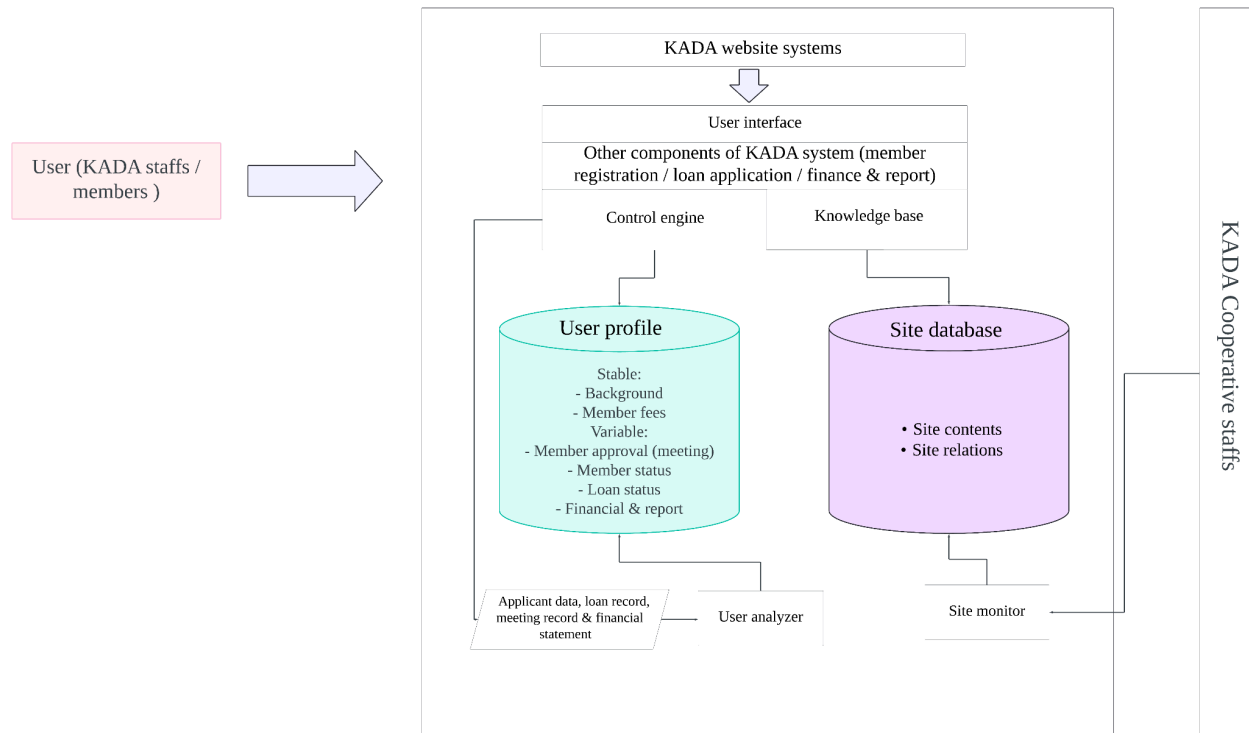
PROJECT ANALYSIS AND DESIGN P3 - GROUP TECH MANIAC

LECTURER : TS. DR. MUHAMMAD IQBAL TARIQ BIN IDRIS

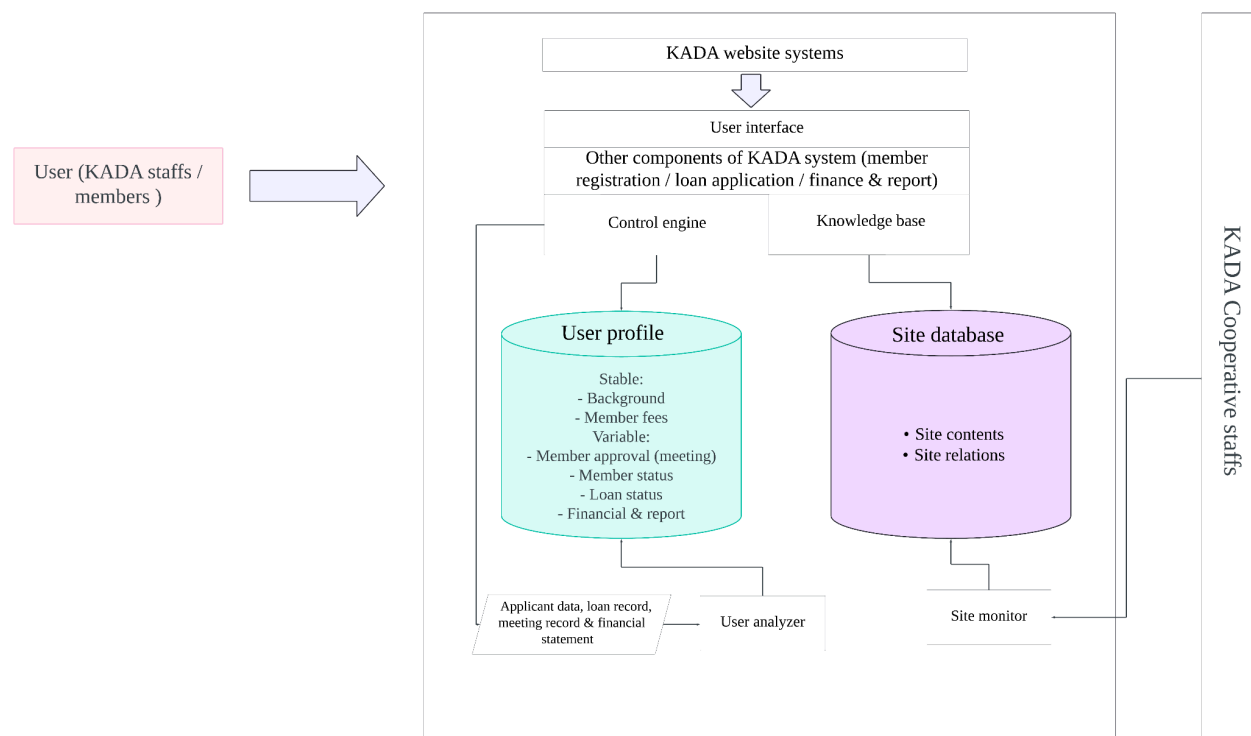
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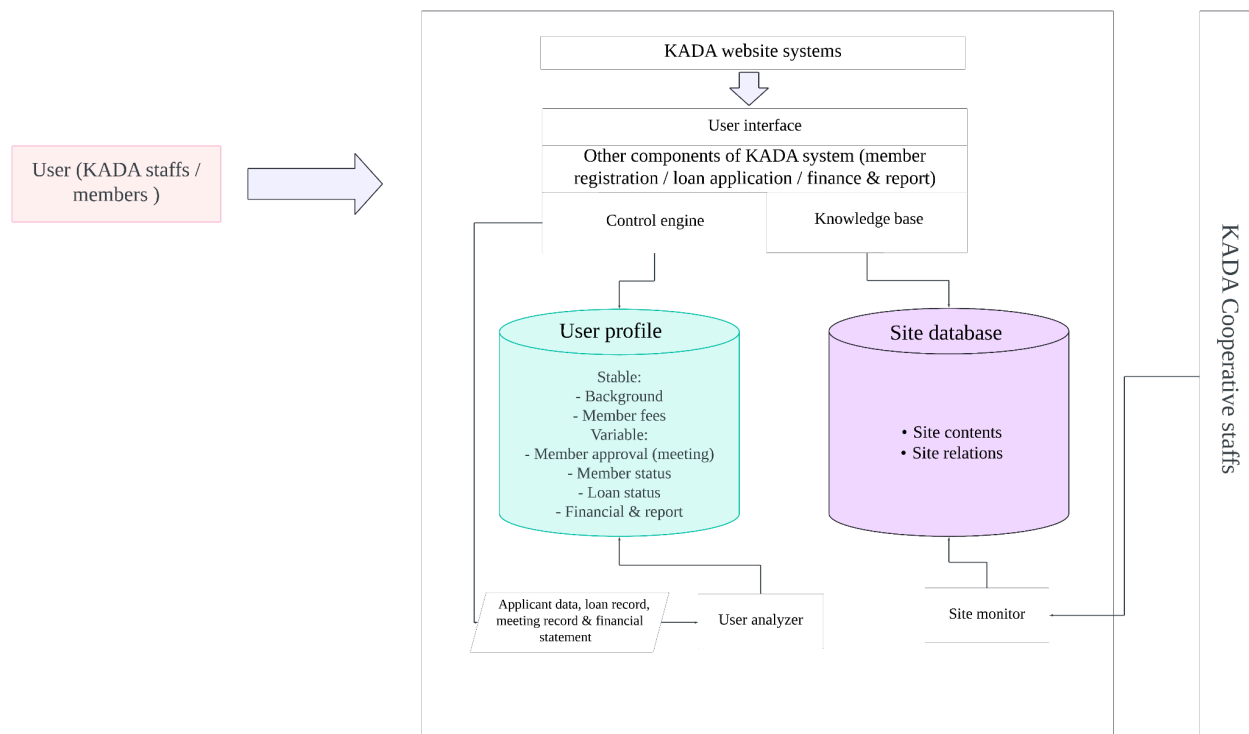
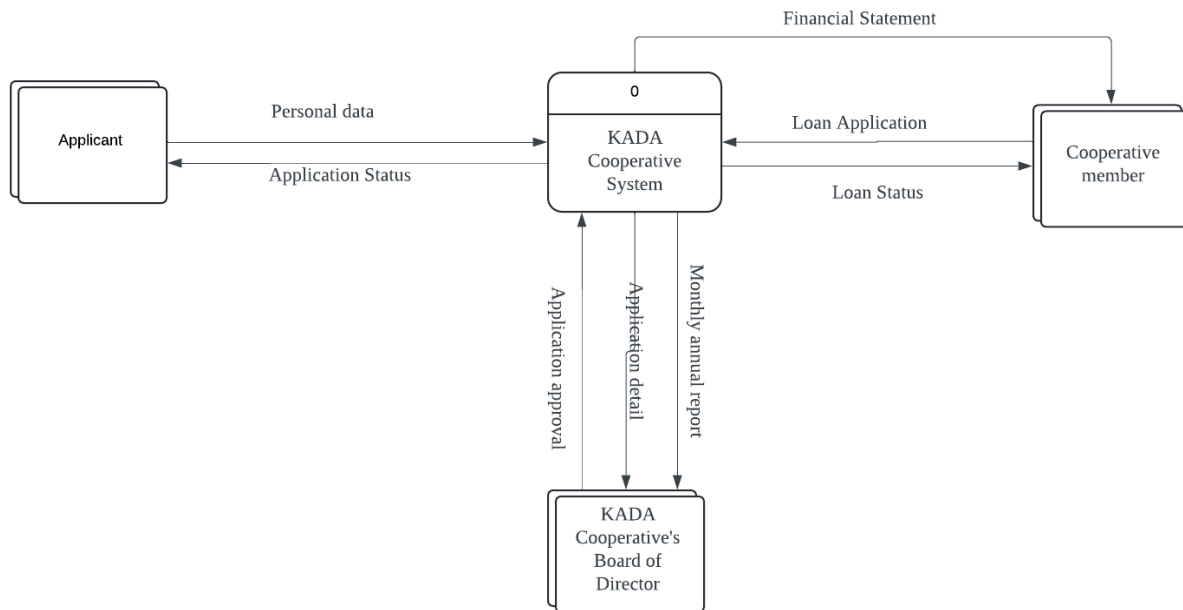


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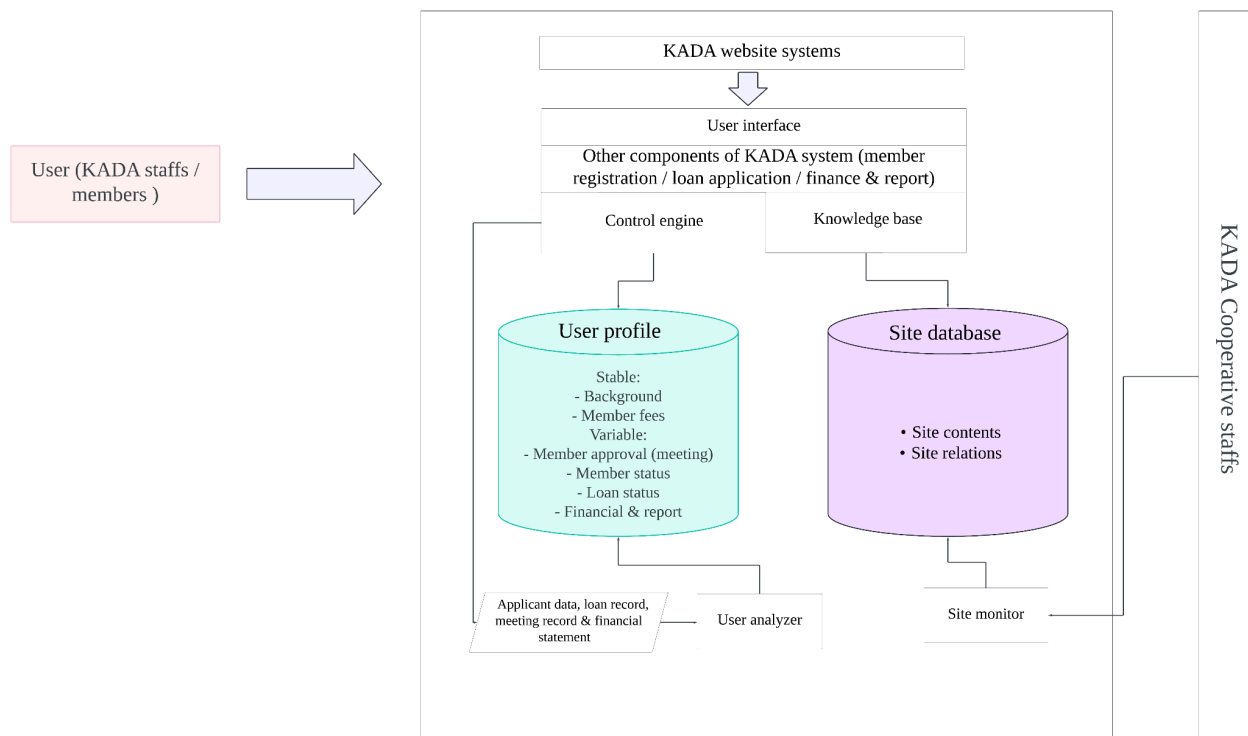


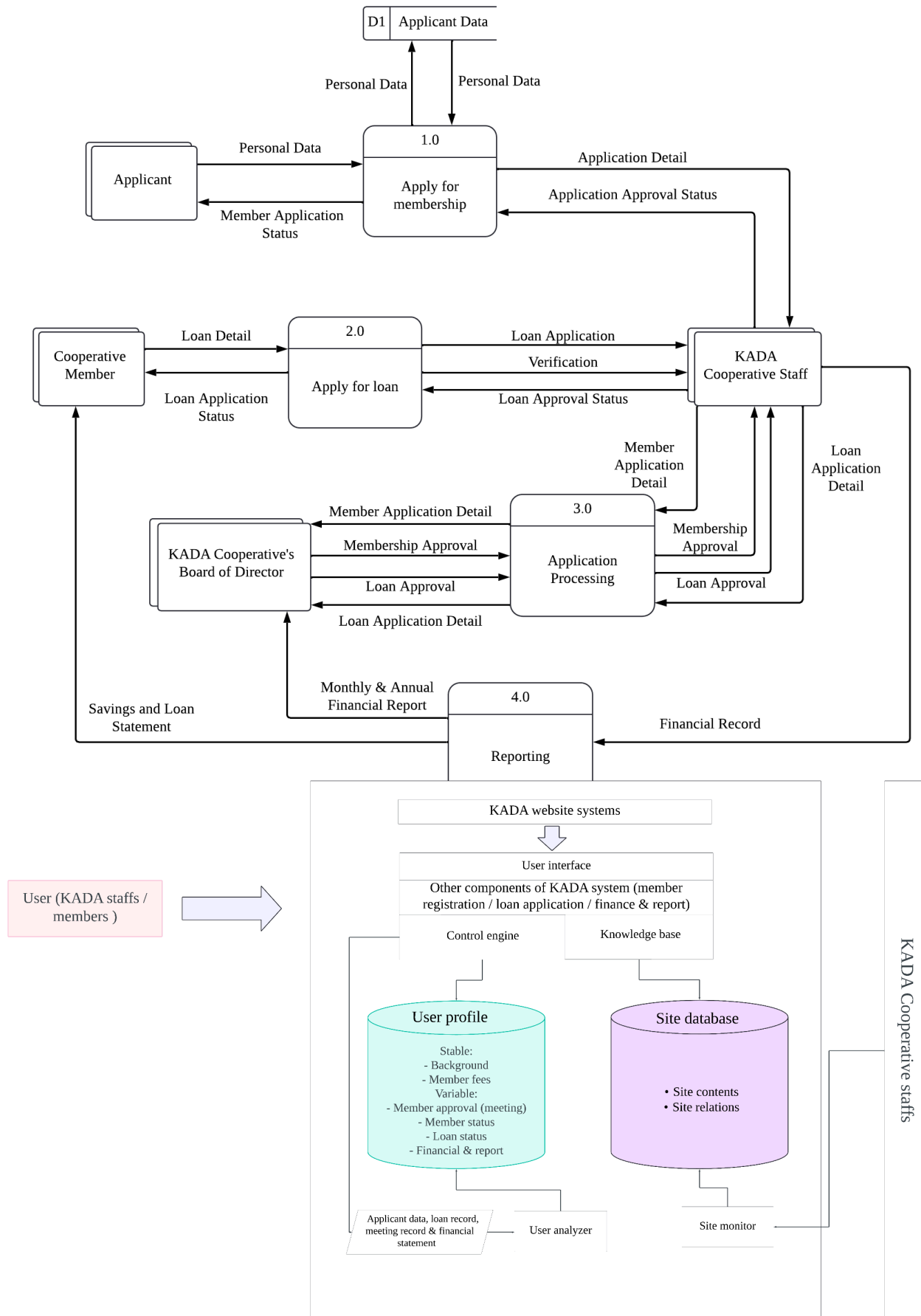
1.0 System Analysis

1.1 Logical DFD Context Diagram

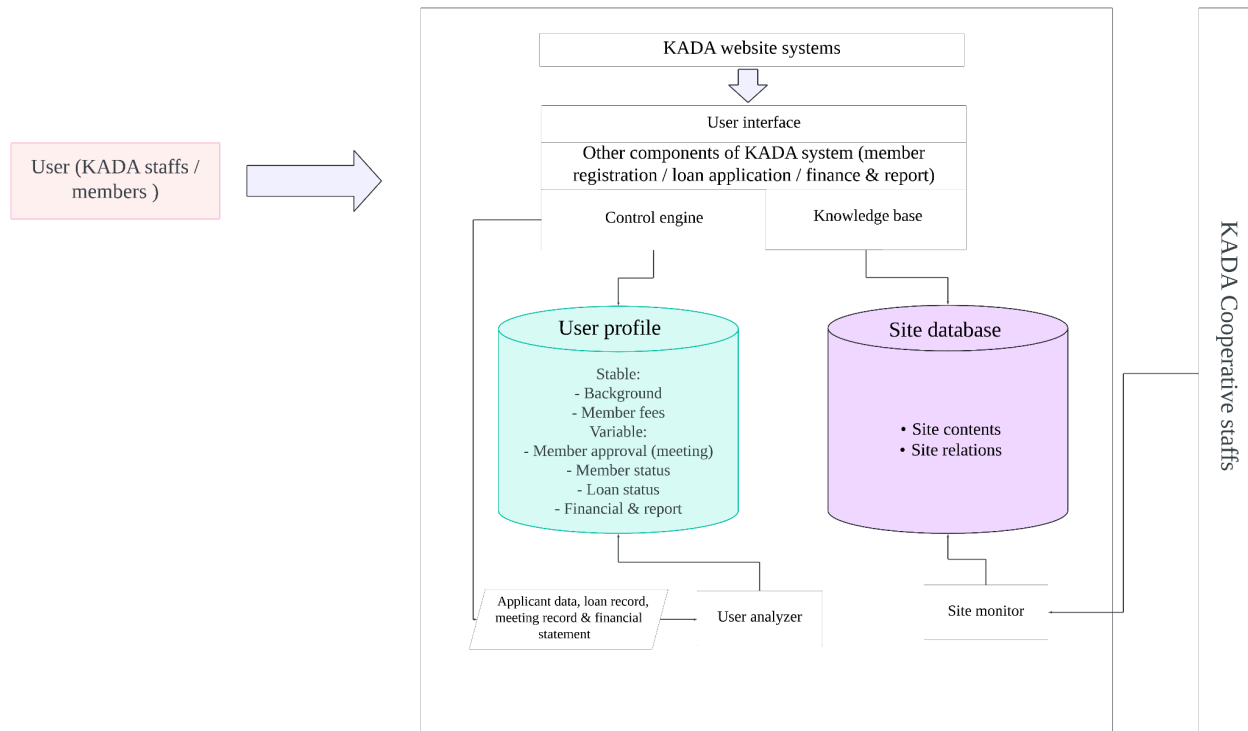
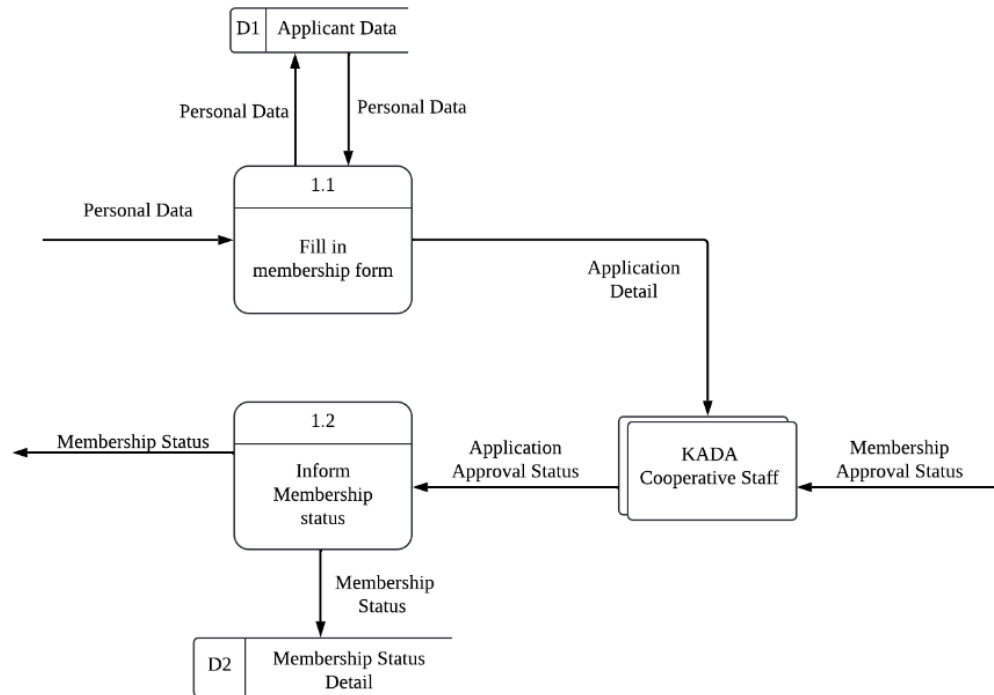


1.2 Logical DFD Diagram Level 0

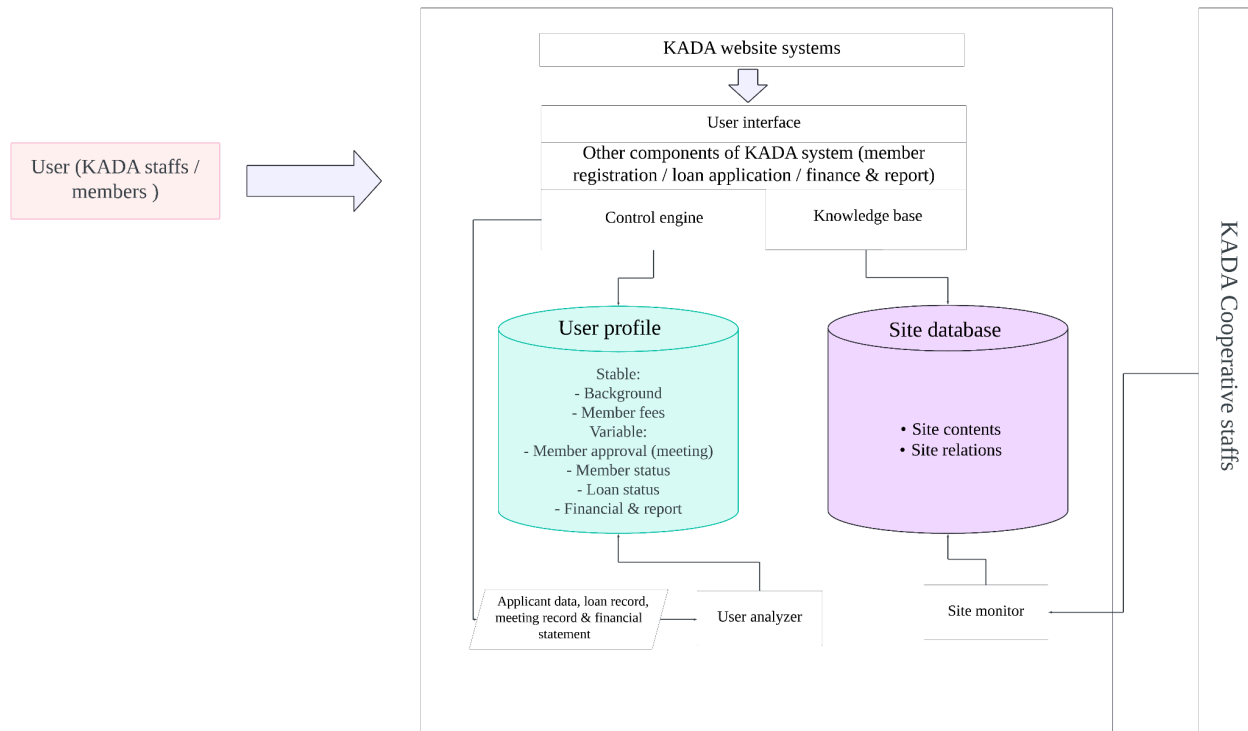
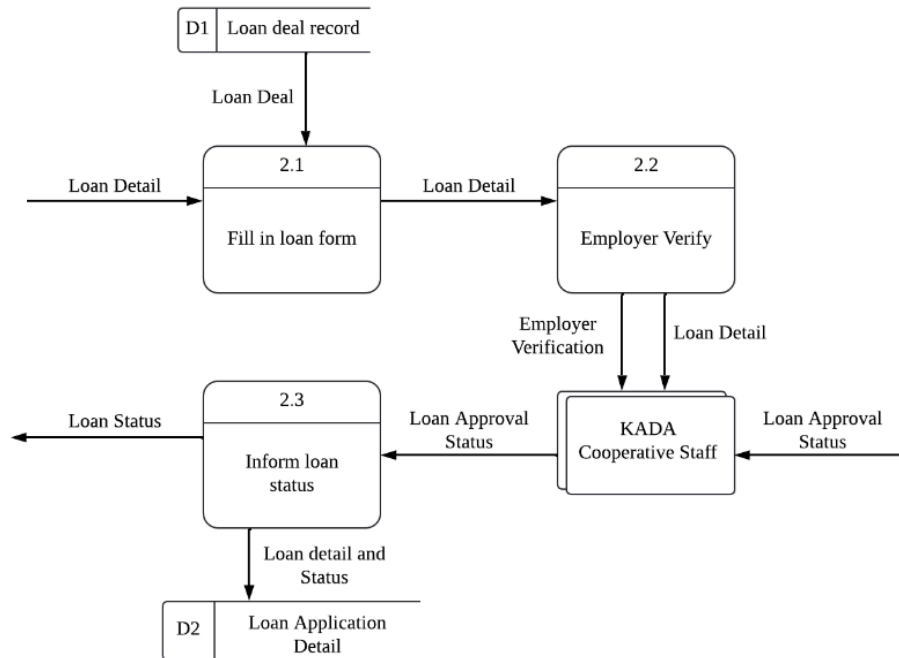




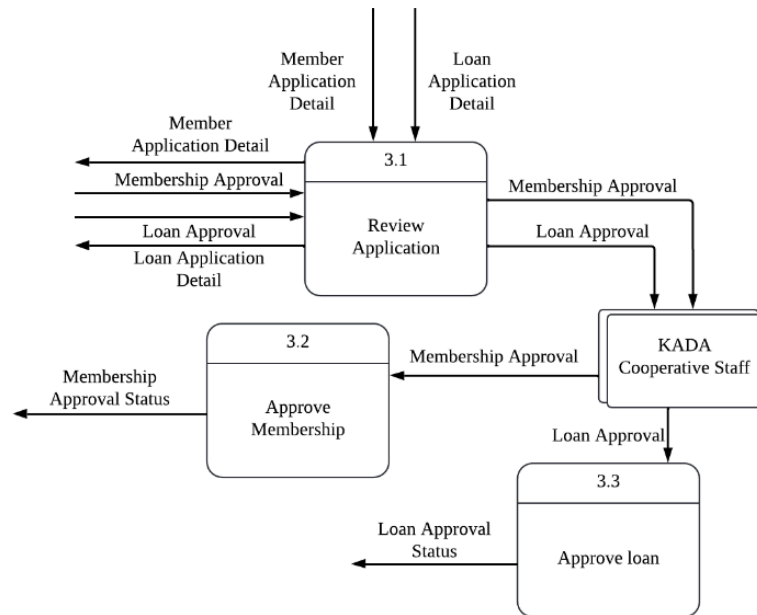
1.3 Logical DFD Child Diagram Process 1



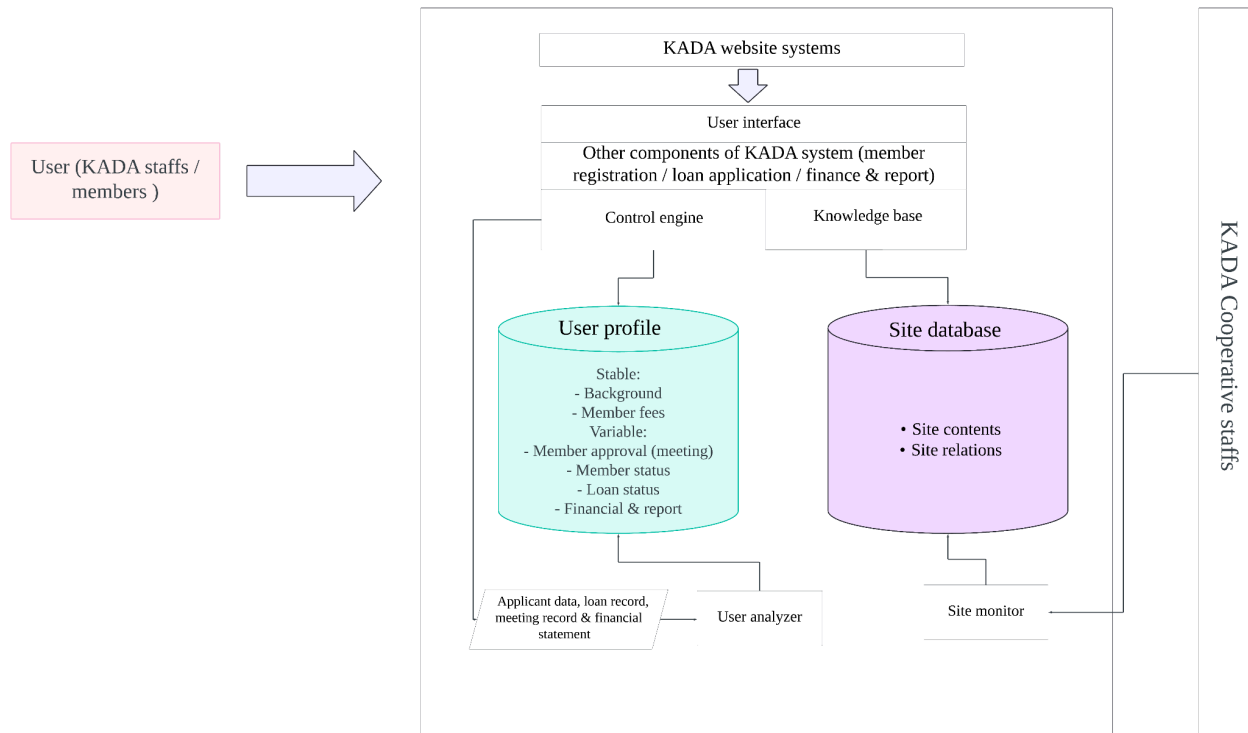
1.4 Logical DFD Child Diagram Process 2

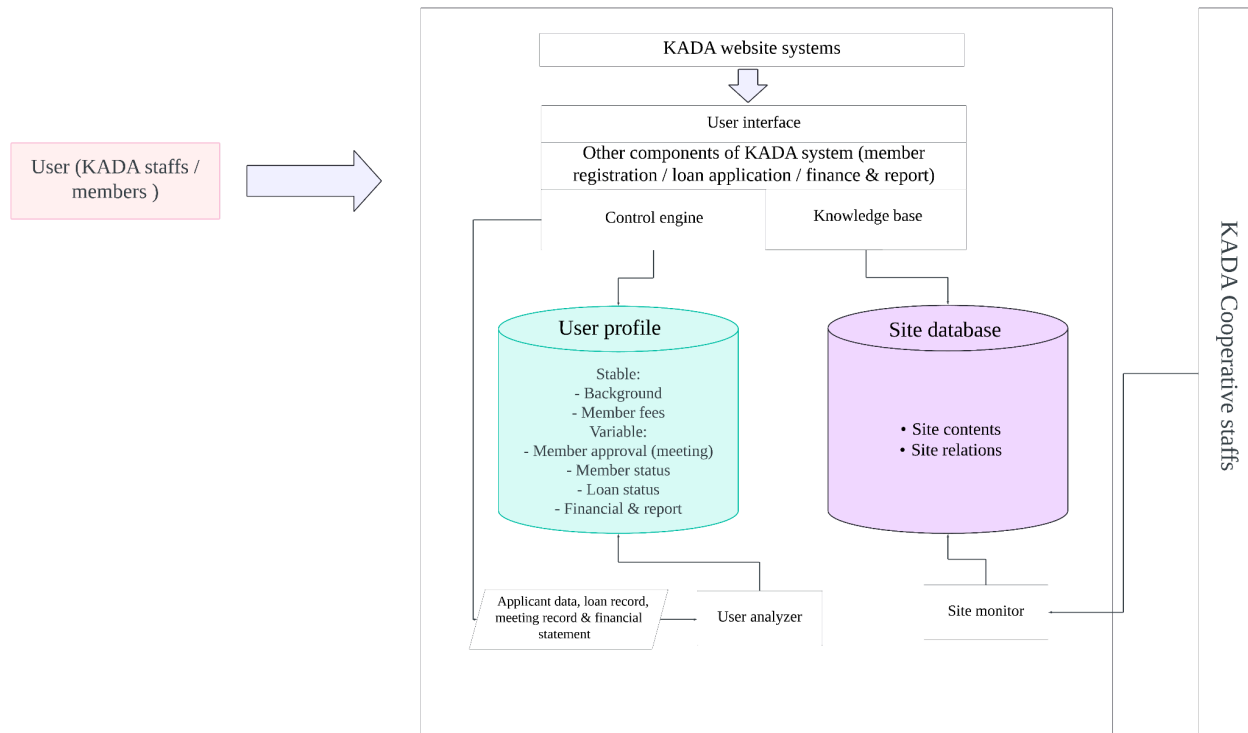
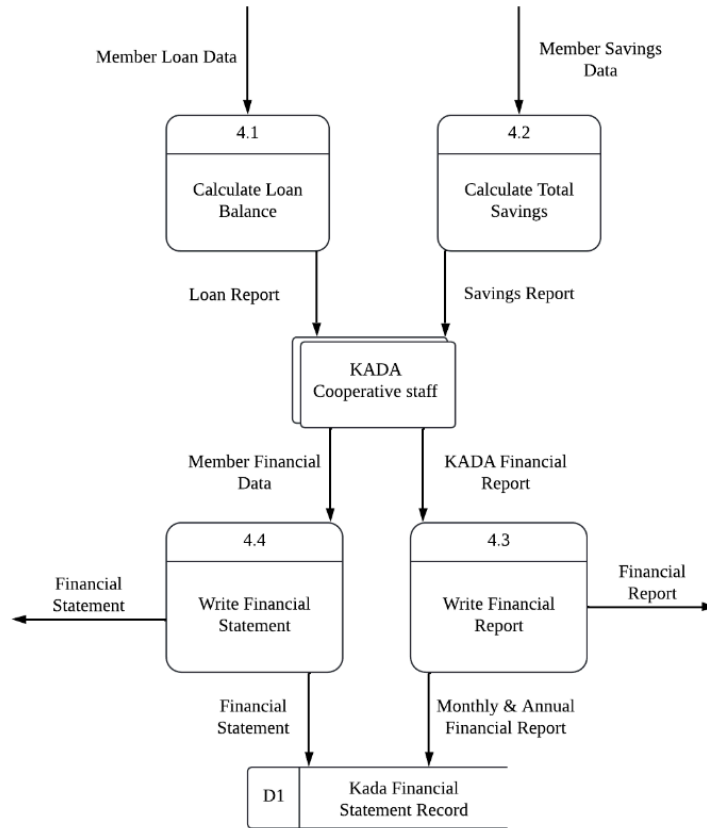


1.5 Logical DFD Child Diagram Process 3



1.6 Logical DFD Child Diagram Process 4





2.0 System Specifications

2.1 Apply For Membership

Process Name: Apply For Membership

Process Name: 1.0

BEGIN

RECEIVE personal details (name, address, contact information) from the applicant.

VALIDATE the format and completeness of the input data.

IF the data is valid THEN

STORE the validated data in the database.

SEND validated data.

GO TO PROCESS 3.0

ELSE

PROMPT the applicant to correct the invalid data.

ENDIF

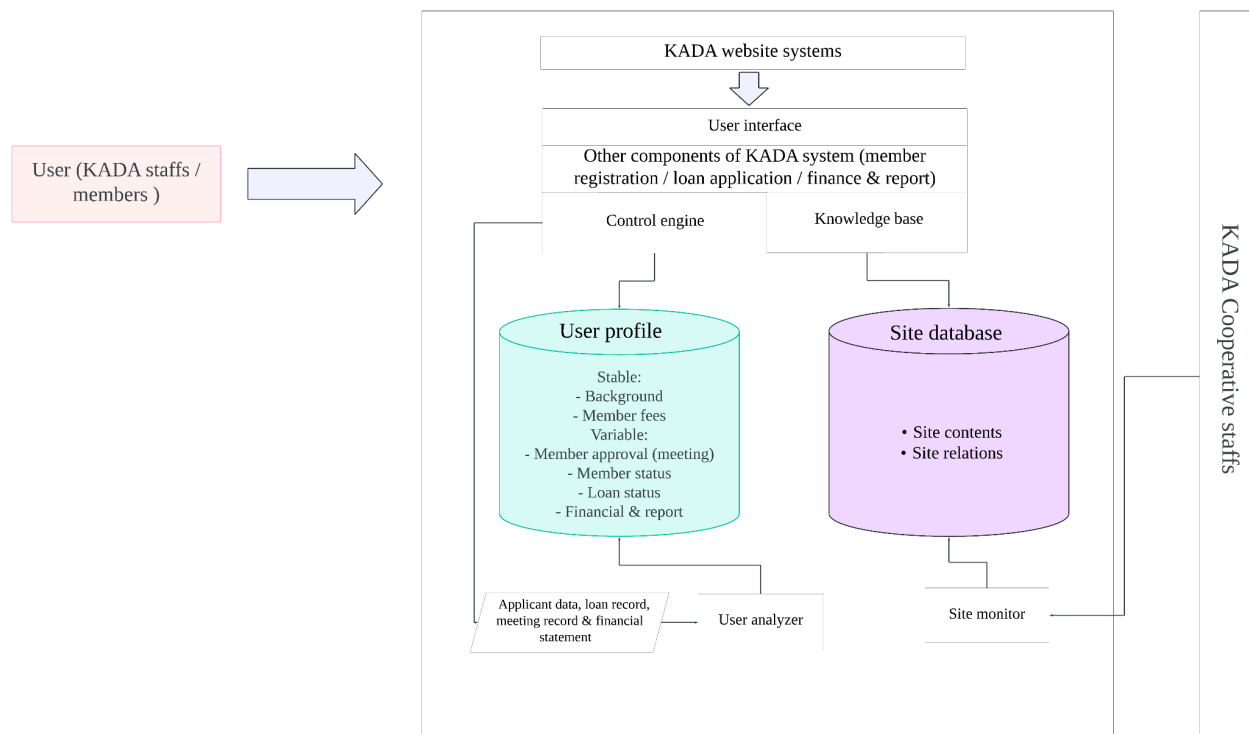
END

2.2 Apply For Loan

Process Name: Apply For Loan

Process Number: 2.0

BEGIN



```

RECEIVE loan application details (personal information, loan amount, purpose, etc.)
from the applicant.
VALIDATE the format and completeness of the input data.
IF the data is valid THEN
    STORE the validated data in the database.
ELSE
    PROMPT the applicant to correct the invalid data.
ENDIF
END

```

2.3 Application Processing

Process Name: Application Processing

Process Number: 3.0

BEGIN

RECEIVE validated personal details from the previous process.

VERIFY eligibility.

IF eligible

CONFIRM status application

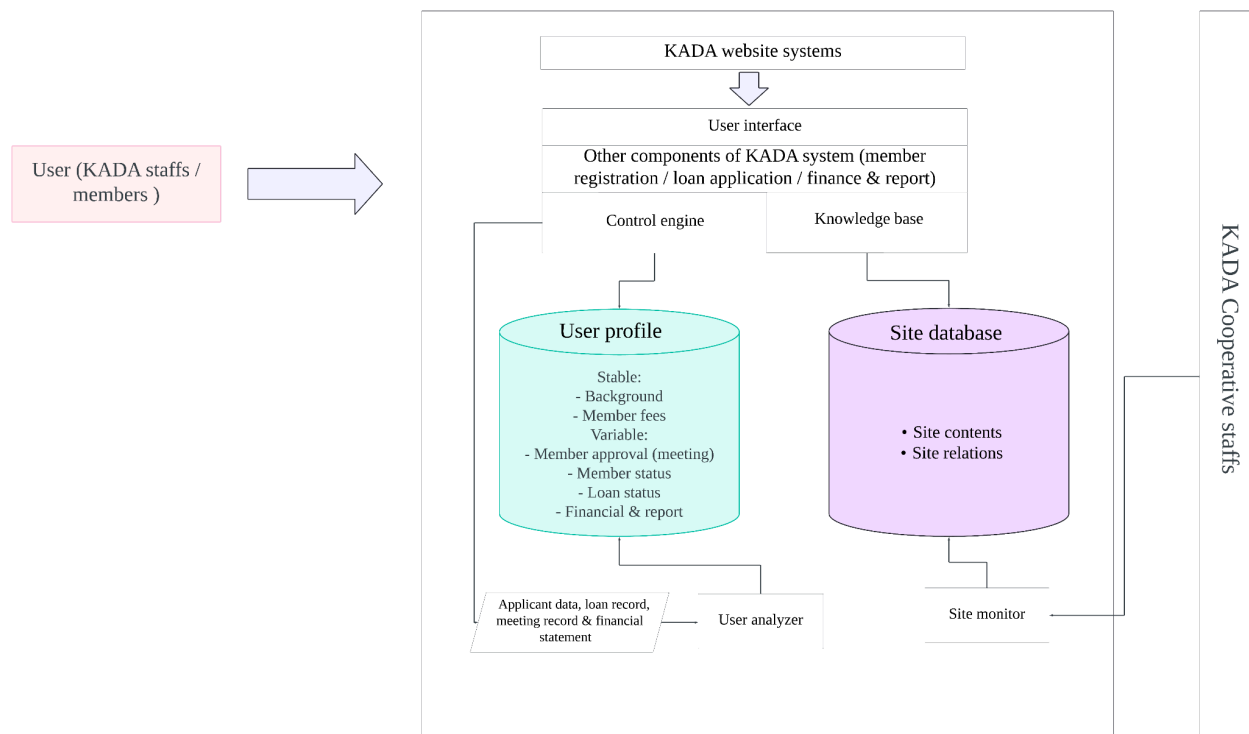
IF membership application

ASSIGN a unique Member ID to the applicant.

ENCRYPT the personal details and the Member ID.

STORE the encrypted data in the system's database.

SEND a membership confirmation email to the applicant.



```

ELSE
    ENCRYPT the personal details and the Member ID.
    STORE the encrypted data in the system's database.
    SEND a loan confirmation email to the applicant.
ENDIF
ELSE
    SEND a rejection email to the applicant.
ENDIF
END

```

2.4 Financial Data Management

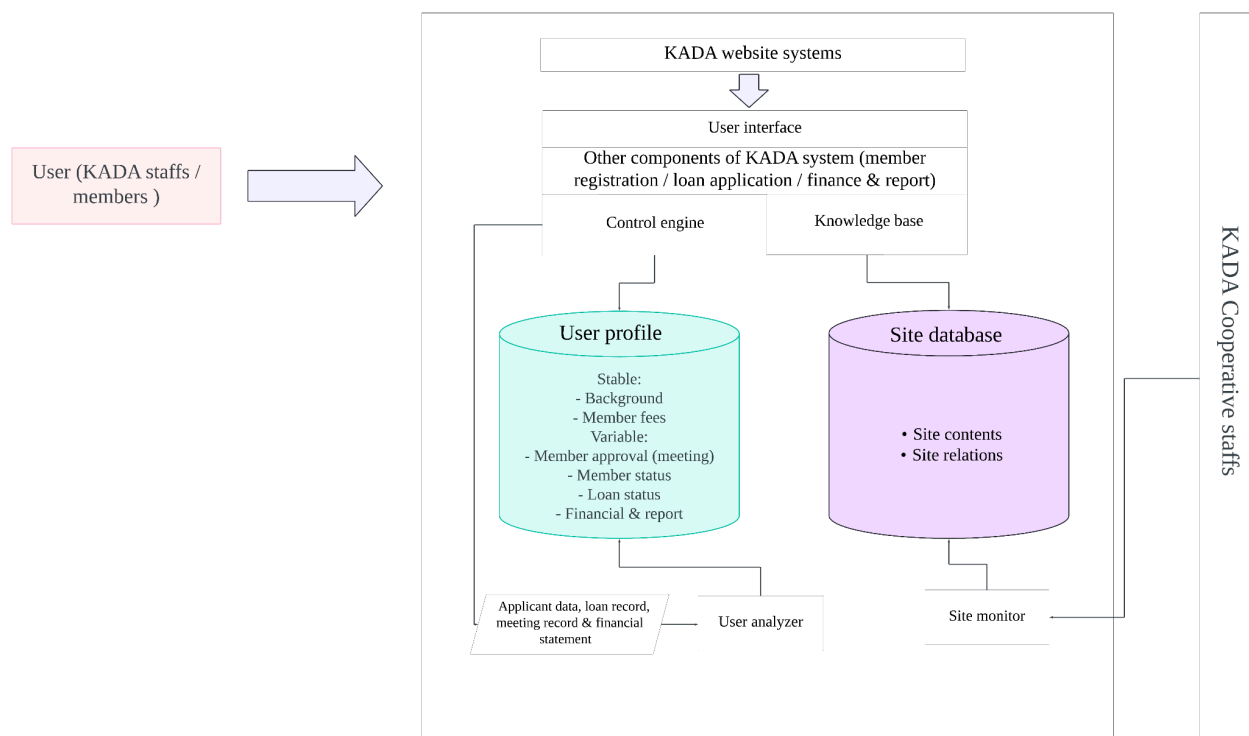
Process Name: Financial Data Management.

Process Number: 4.0

```

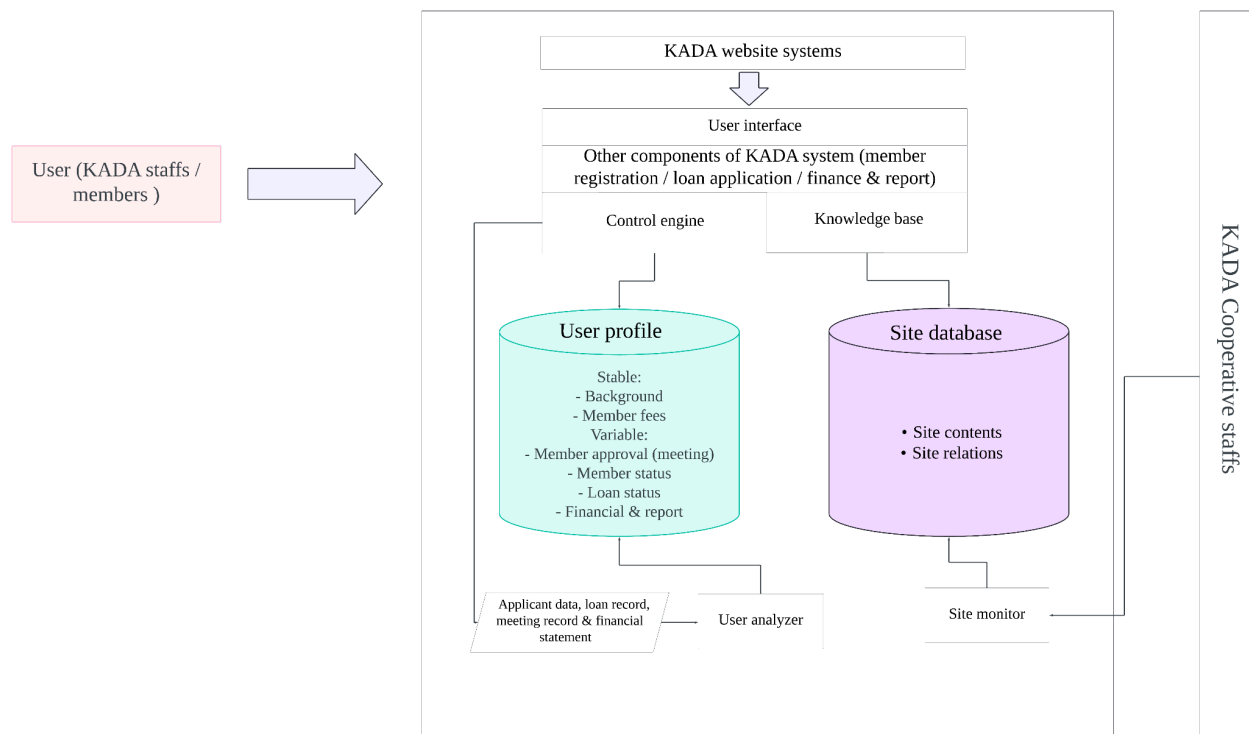
BEGIN
    RECEIVE Member Loan Data and Member Savings Data.
    CALCULATE loan balances and total savings.
    GENERATE financial reports and statements.
    STORE financial reports and statements in the system.
    SEND financial reports to stakeholders and financial statements to members.
END

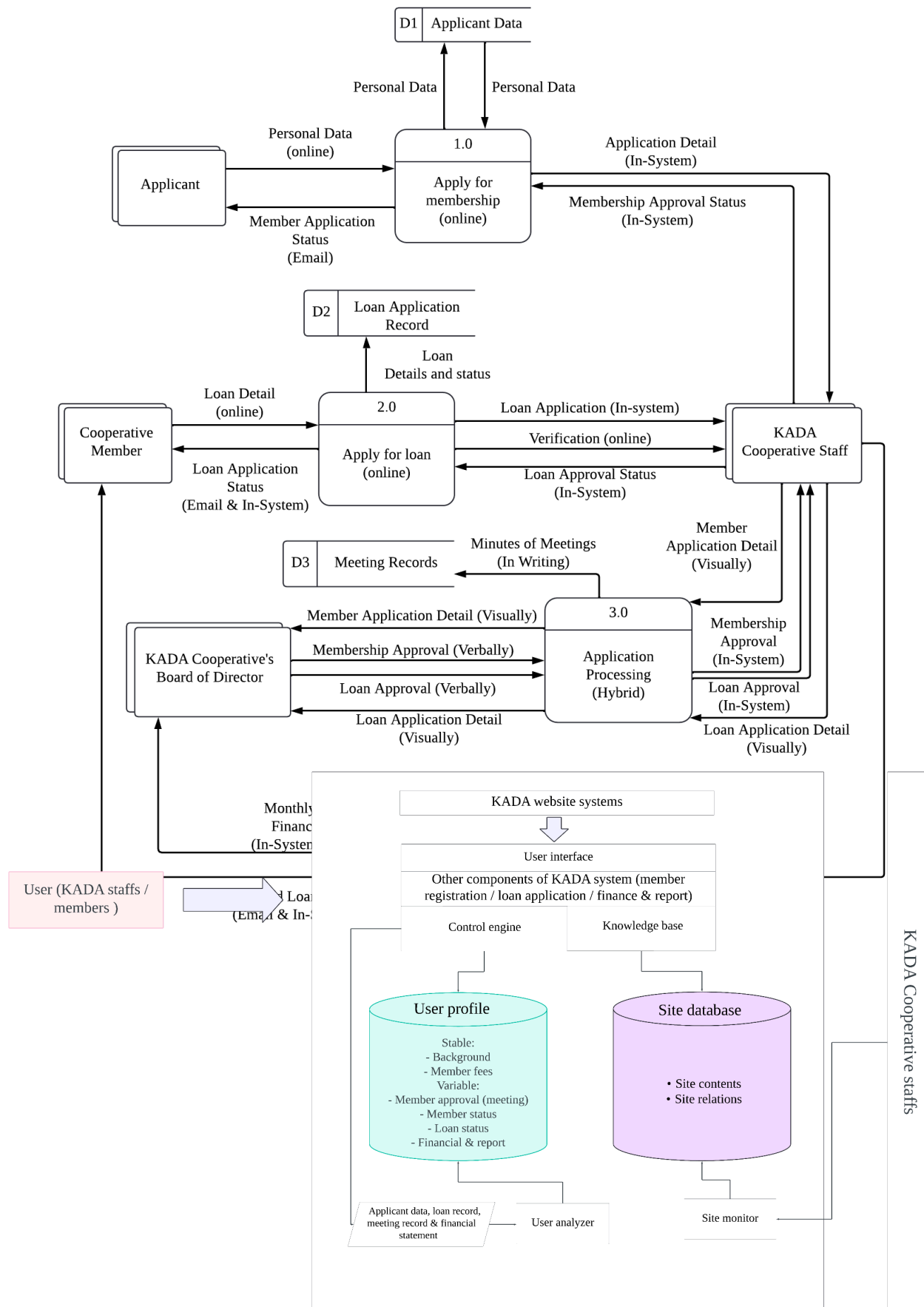
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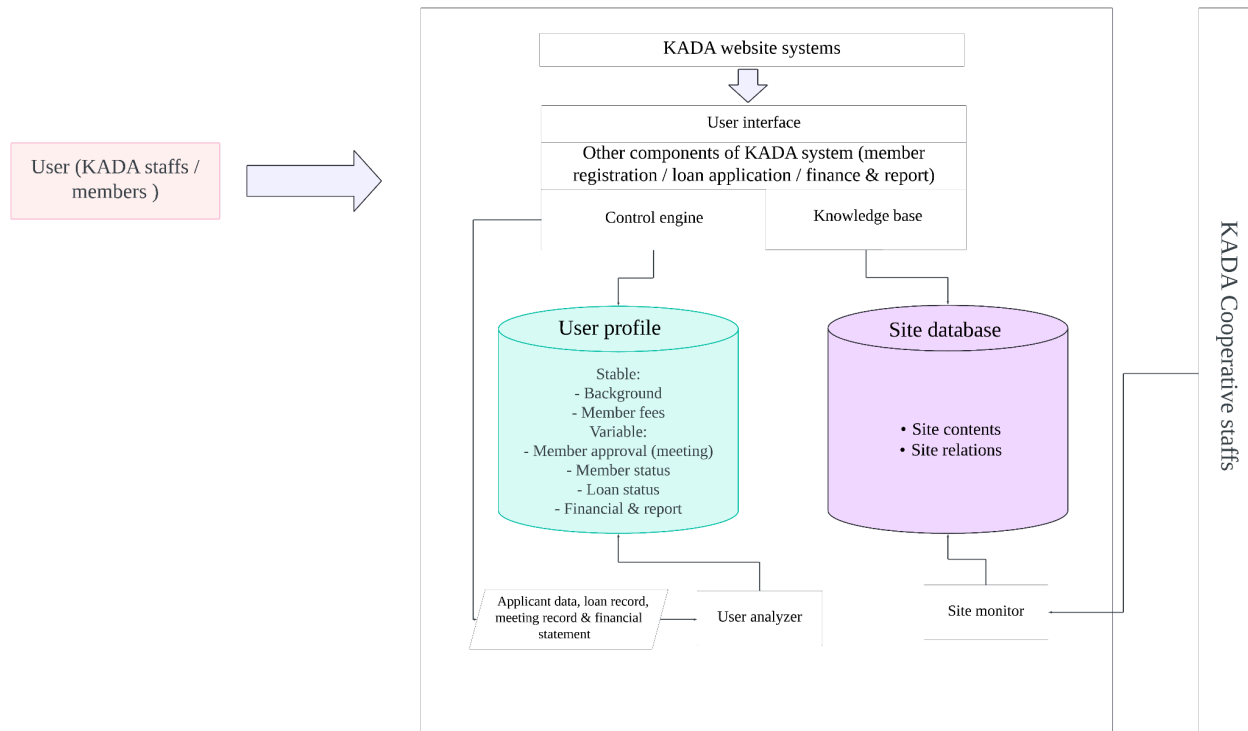
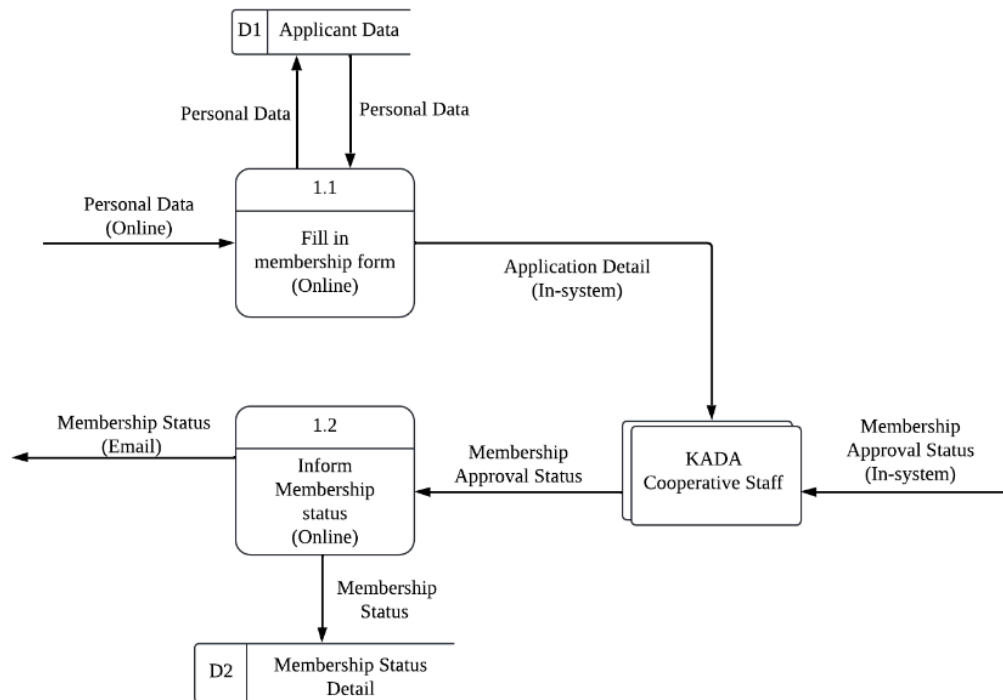
3.0 System Design

3.1 Physical DFD Level 0 Diagram

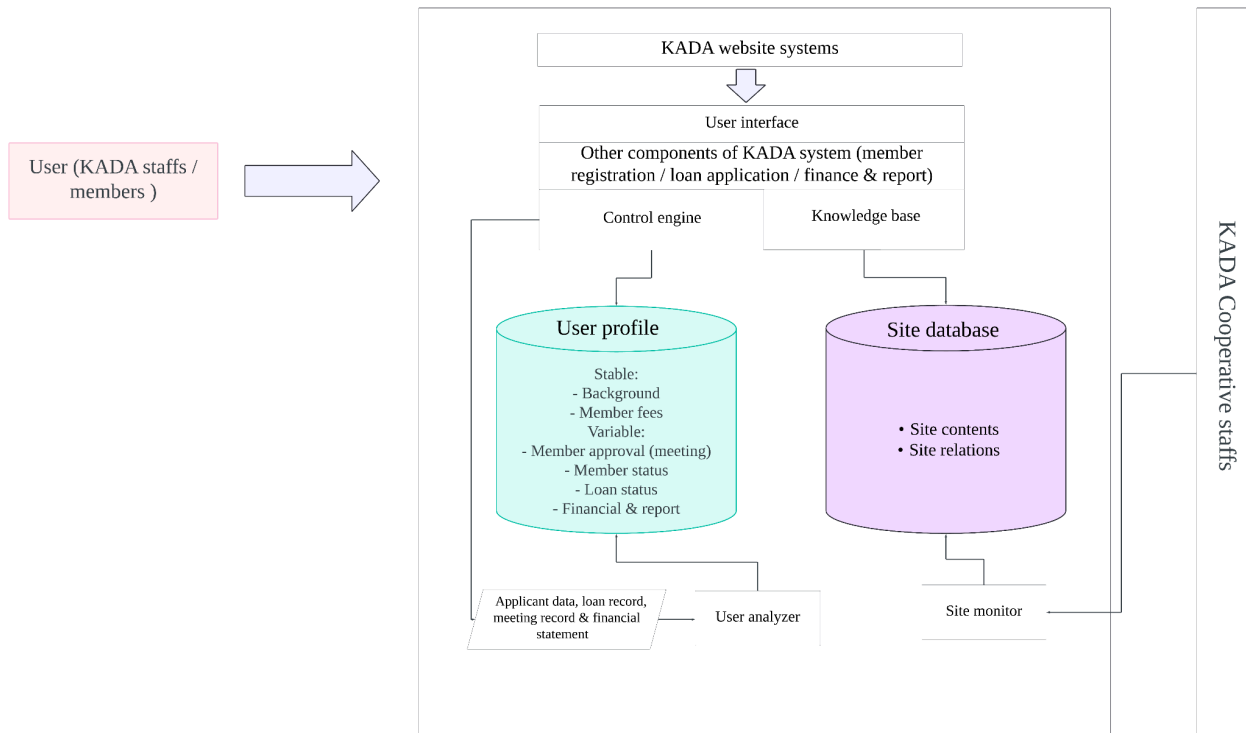
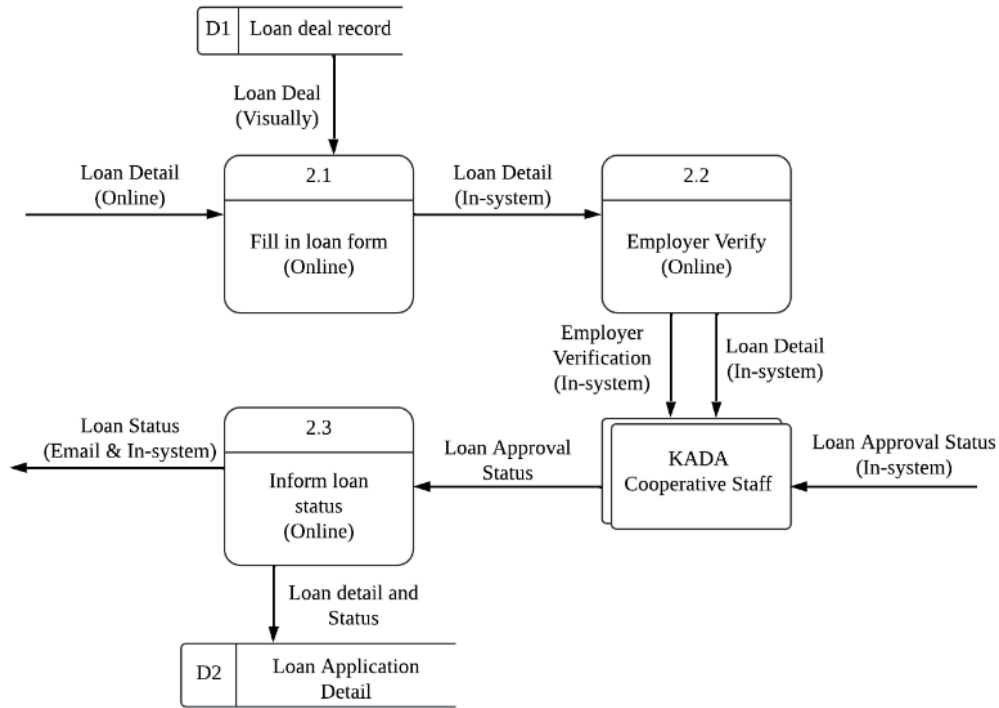




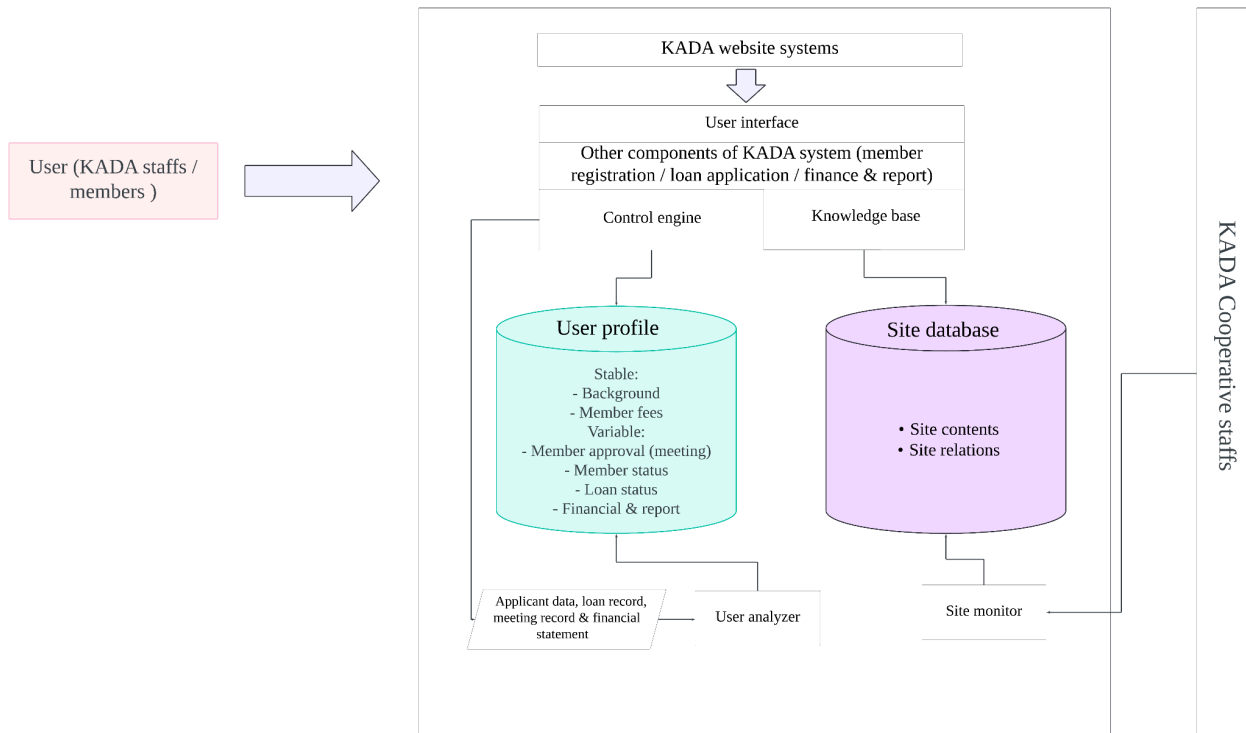
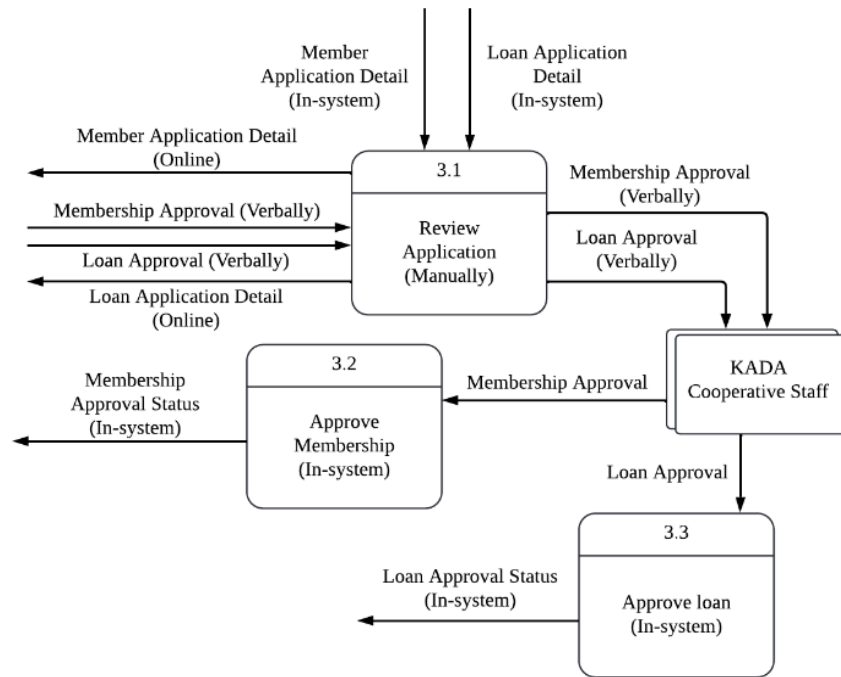
3.2 Physical DFD Child Diagram Process 1.0



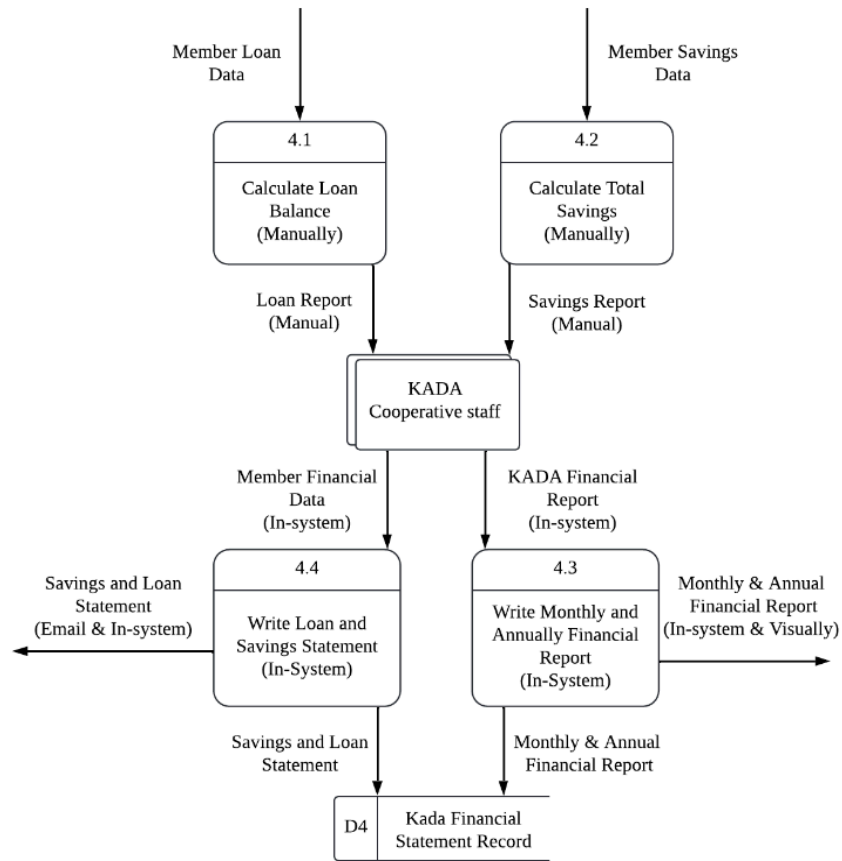
3.3 Physical DFD Child Diagram Process 2.0



3.4 Physical DFD Child Diagram Process 3.0

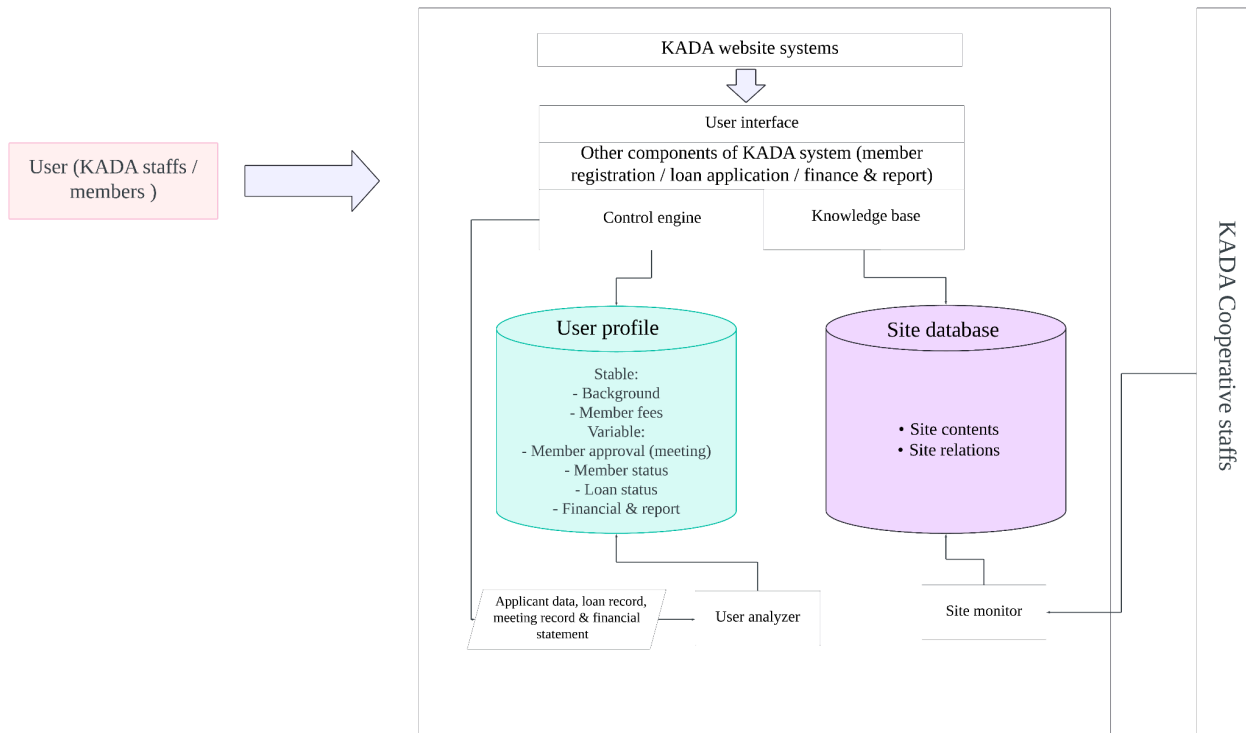


3.5 Physical DFD Child Diagram Process 4.0



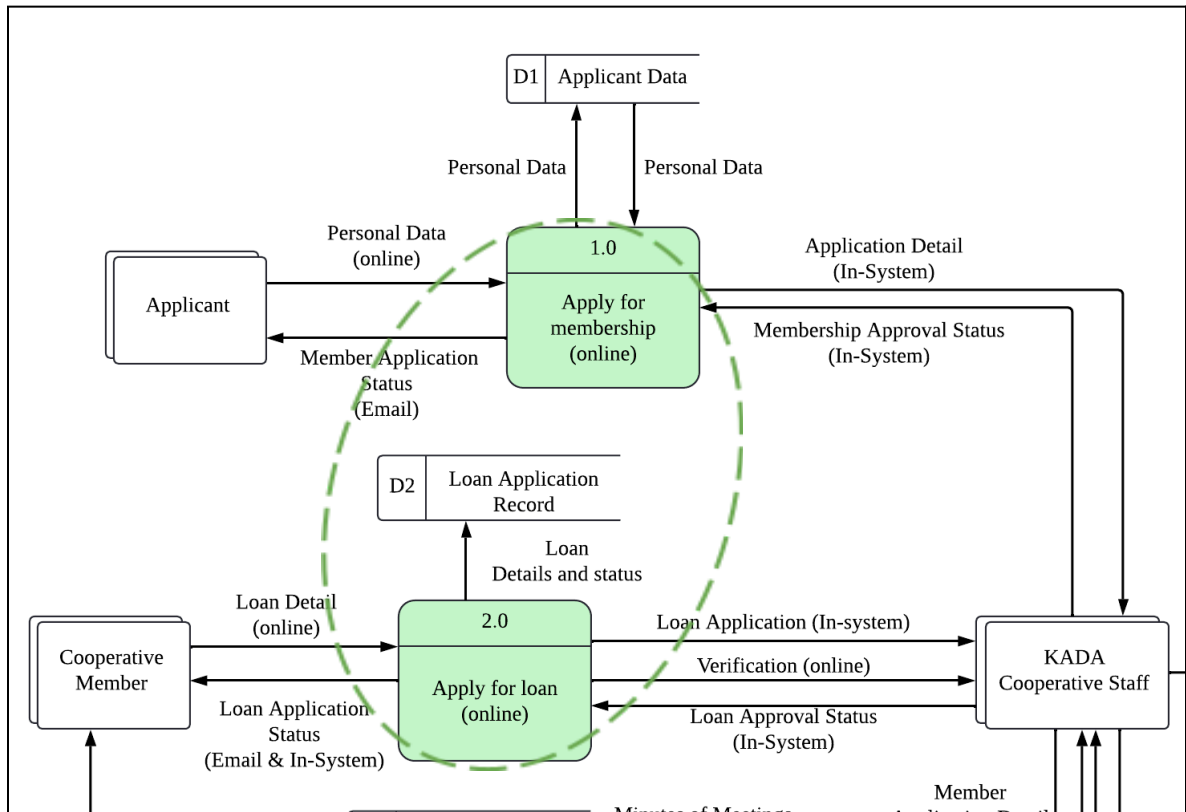
3.6 Partitioning

- Group 1 :

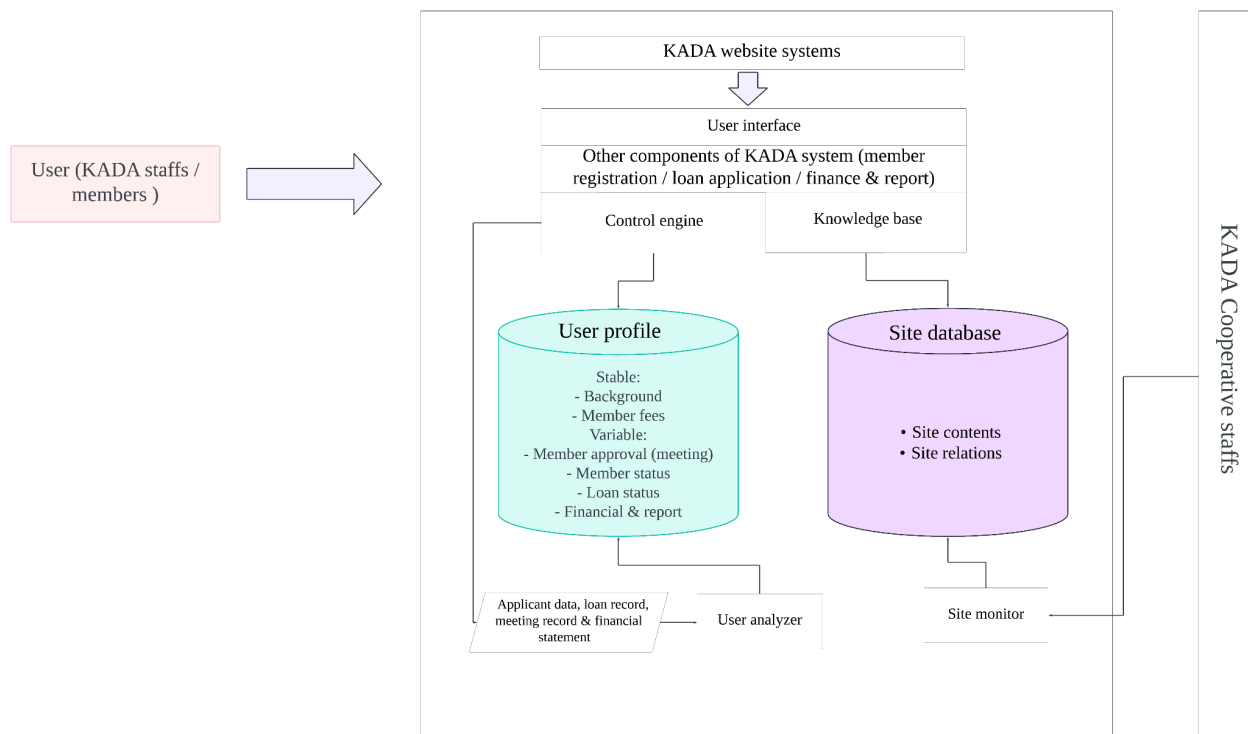


Process 1.0 and Process 2.0 : Similar Task

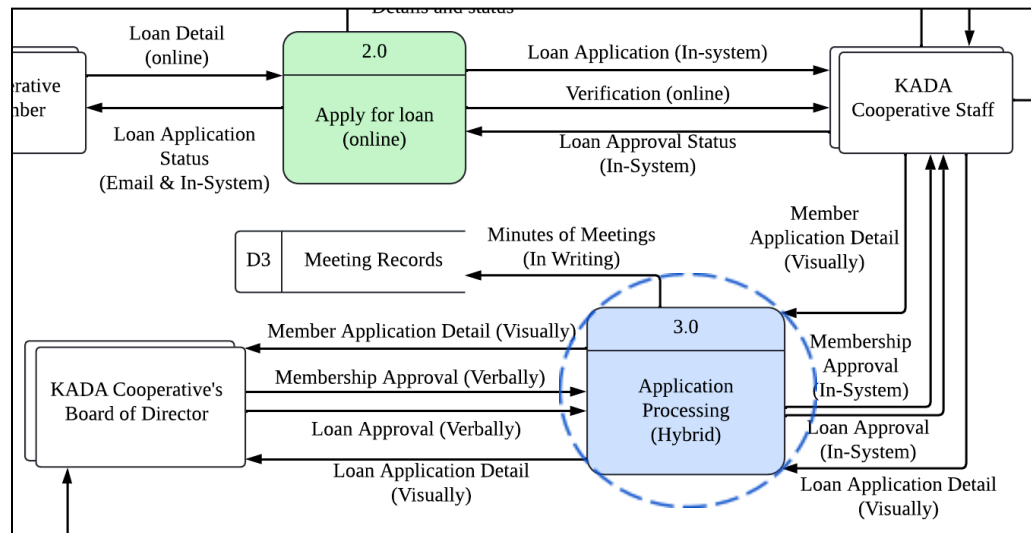
Indicates processes that collect user information and update the application status.



- Group 2 :
Process 3.0 : Different Users



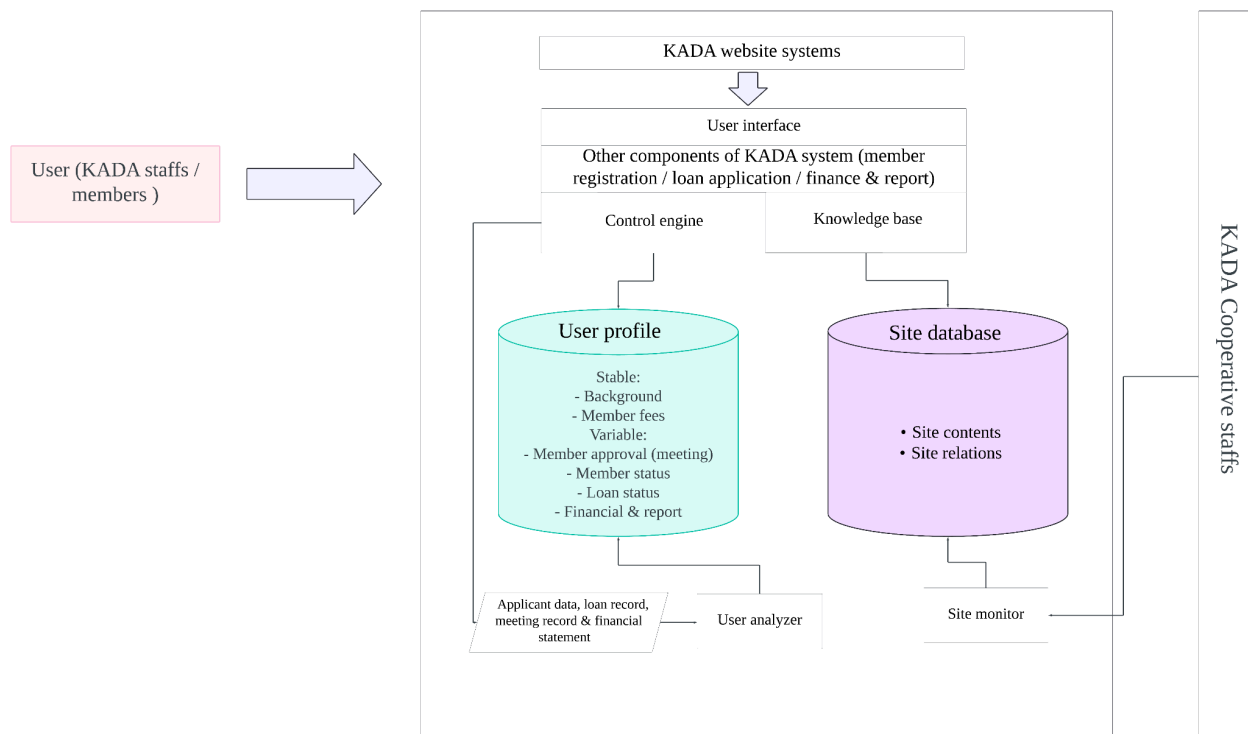
Indicates processes that verifying member application and loan applications through a meeting discussion between board of directors and system admin.

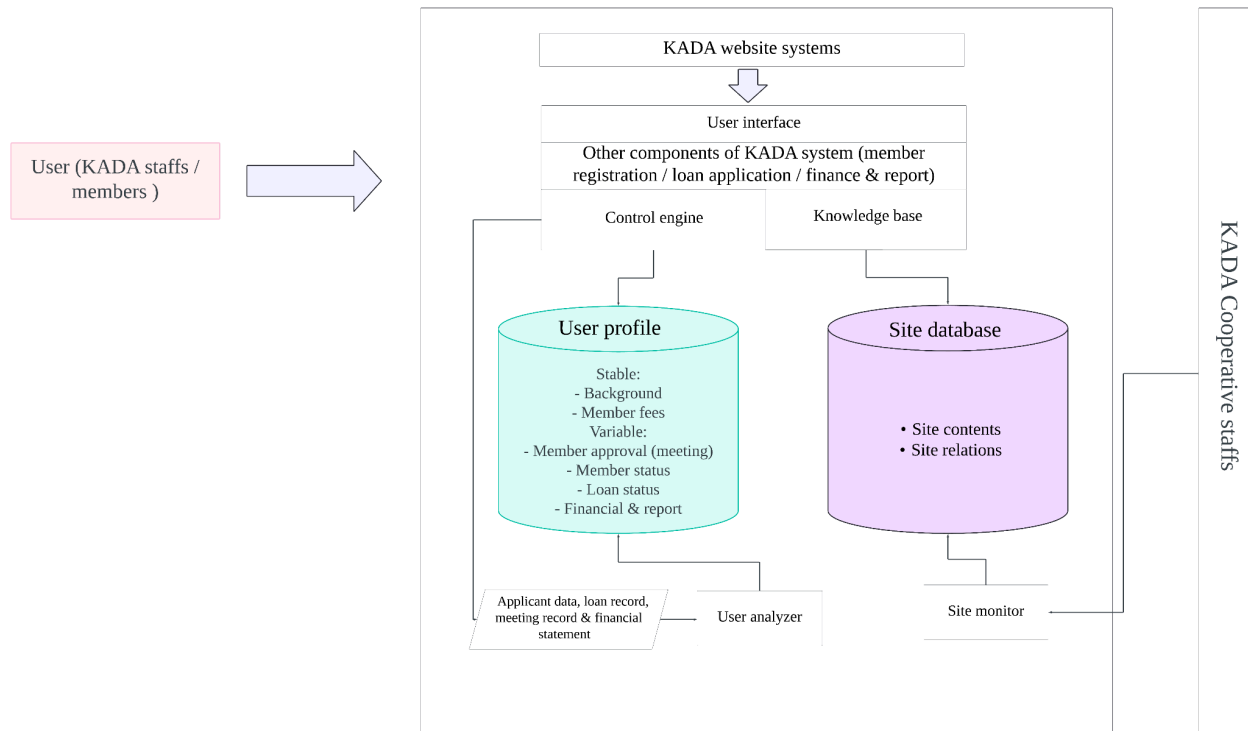
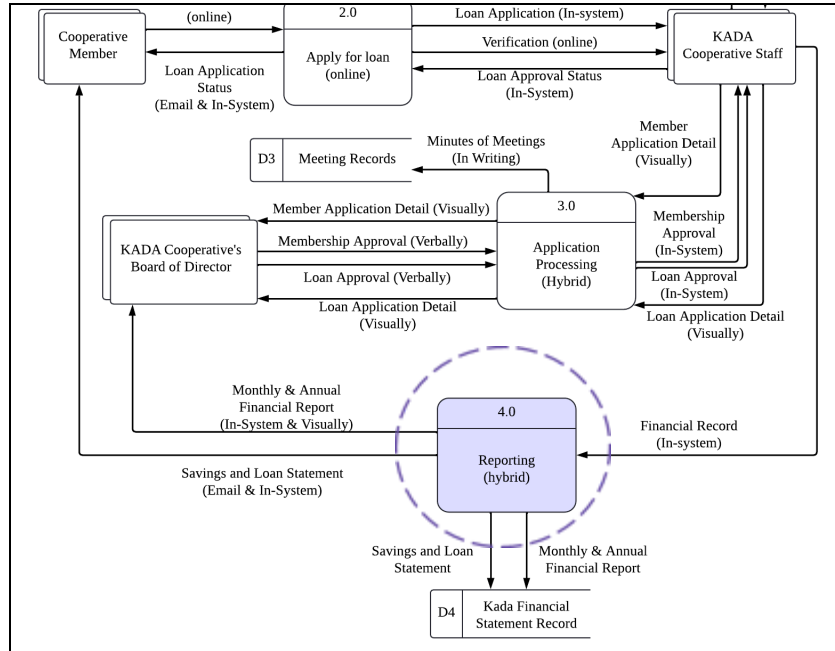


- Group 3 :

Process 4.0 : Batch Process

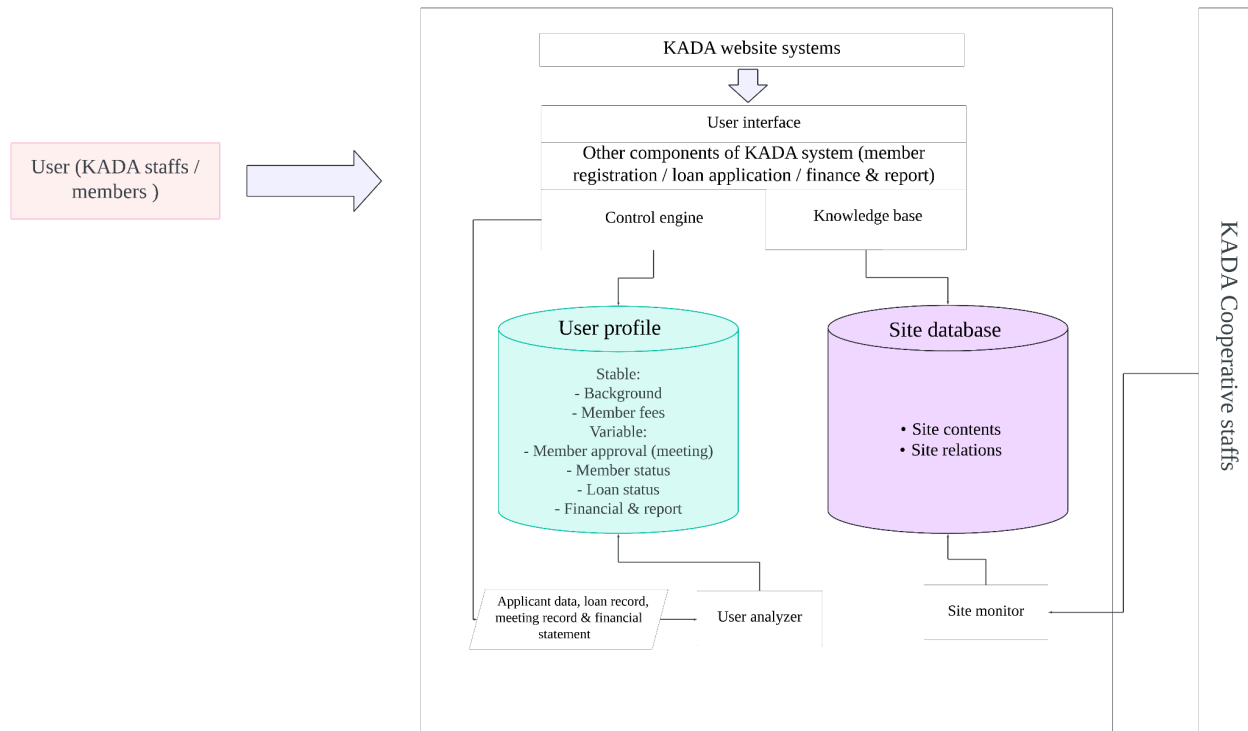
Indicates process reporting financial statements and reports from the KADA system to Cooperative Member and KADA board of directors.



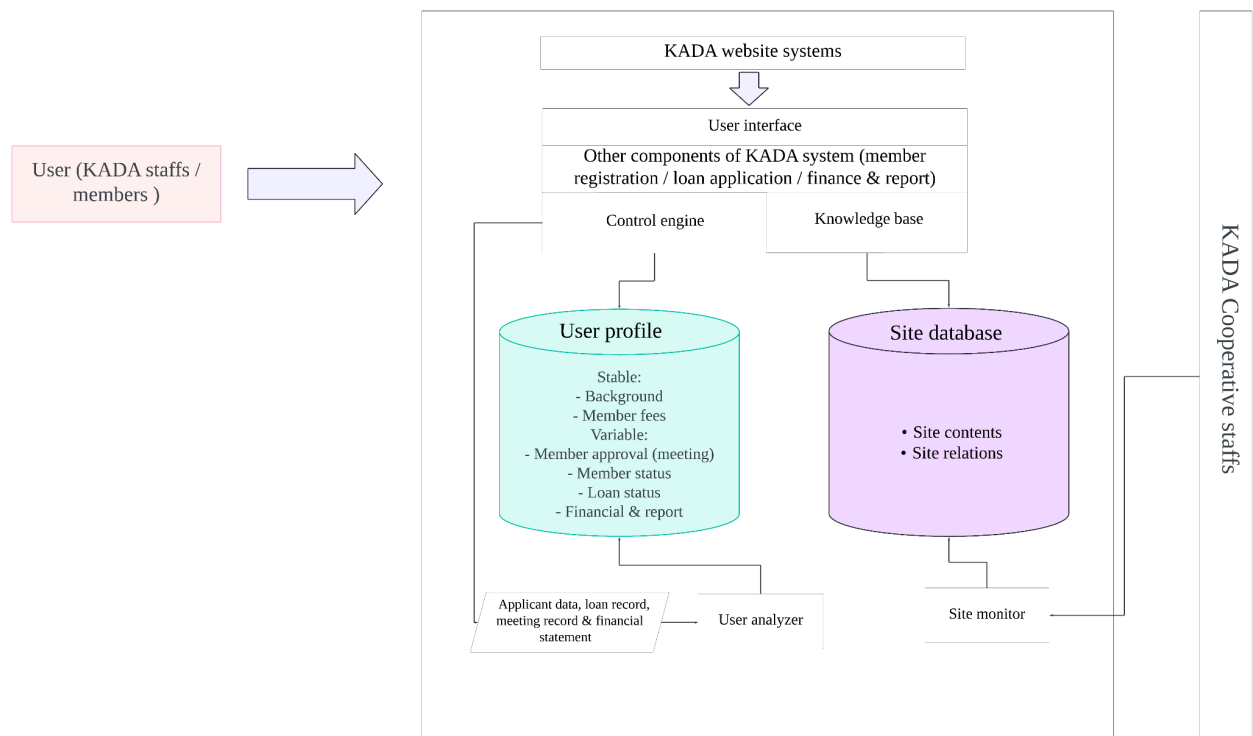


3.4 CRUD Matrix

Activity	Applicant Data	Meeting Record	Membership Status Detail	Loan Deal Record	Loan Application Detail	KADA Financial Statement Record
Fill in membership form	CR					
Inform membership status	R		CU			
Review application		RU				
Approve membership	R	U	R			
Fill in loan application form		R		R	CR	
Inform loan status					U	
Approve loan	R	U			R	
Calculate loan balance				R		RU
Calculate total savings						RU
Write loan and savings statement						CU
Write monthly and annually financial report						CU

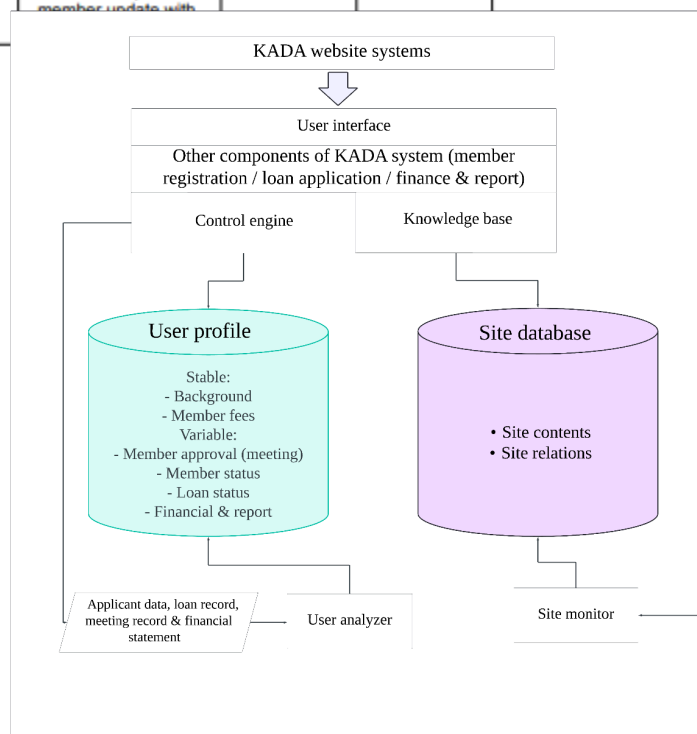
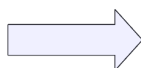


3.5 Event Response Table



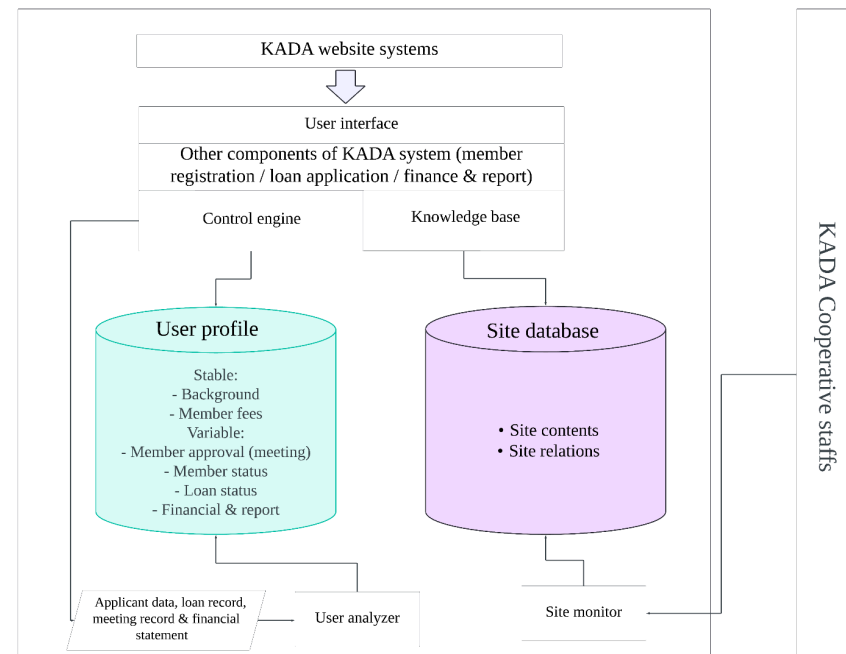
Event	Source	Trigger	Activity	Response	Destination
Register to be a member of KADA	Applicant	Personal data	Applicants need to fill their personal information into the application form to be a member of KADA and system admin will announce their status	Member application status	Applicant
Apply for loan	Cooperative member	Loan detail	Member should fill in their loan detail and wait for the status approval by KADA cooperative staffs	Loan application status	Cooperative member
Obtain application status	KADA Cooperative Staff	Application status	KADA Cooperative Staff will get approval of member applications and loan applications via board of director meeting. Notify the applicants through the system.	Application detail	KADA Cooperative Staff
Review member application detail	KADA Cooperative's Board of Director	Membership approval	KADA Cooperative's Board of Director will approve the applications that qualify to be a cooperative member of KADA through the meeting session.	Member application detail	KADA Cooperative's Board of Director
Approve the loan application	KADA Cooperative's Board of Director	Loan approval	KADA Cooperative's Board of Director supposed to announce the qualified loan application through the meetings for KADA Cooperative Staff to announce it in the system.	Loan application detail	KADA Cooperative's Board of Director
Insert the financial report	KADA Cooperative Staff	Financial record	KADA Cooperative Staff required to summary the financial reports monthly and annually for the KADA Cooperative's Board of Director to review.	Monthly and Annual report	KADA Cooperative's Board of Director
Savings and loan statement for cooperative member to review	KADA Cooperative Staff	Financial record	KADA Cooperative Staff will update the cooperative members savings and loan statement in the system to let the cooperative member update with	Savings and loan statement	Cooperative member

User (KADA staffs / members)

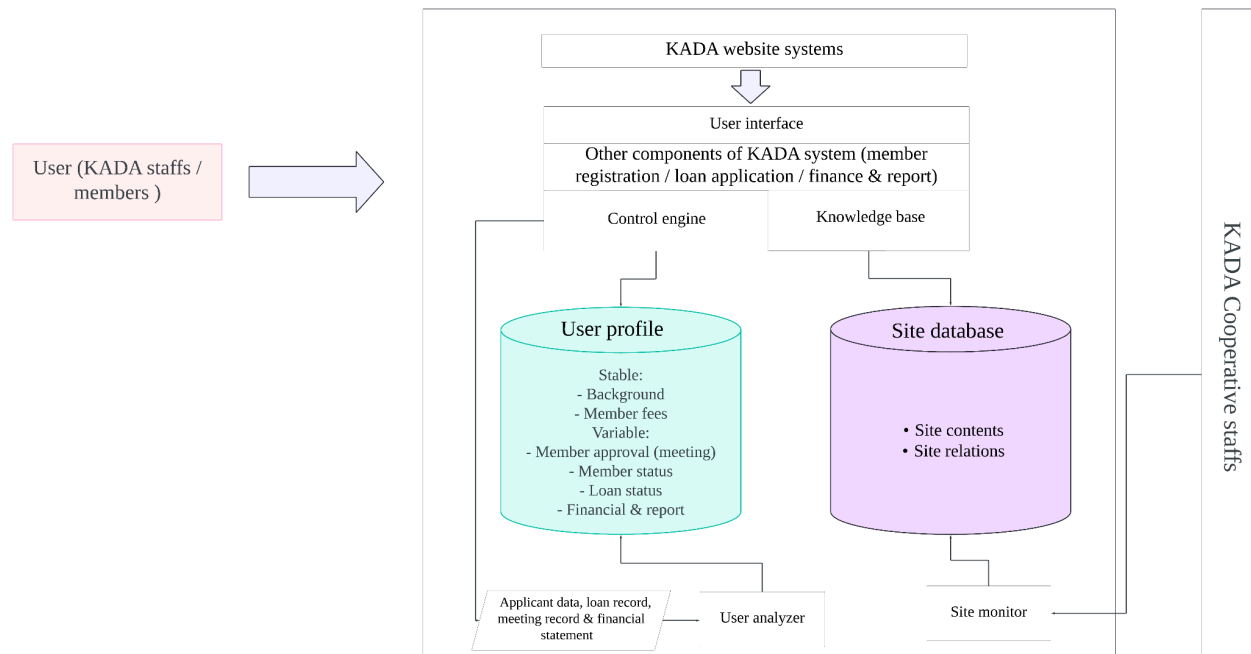


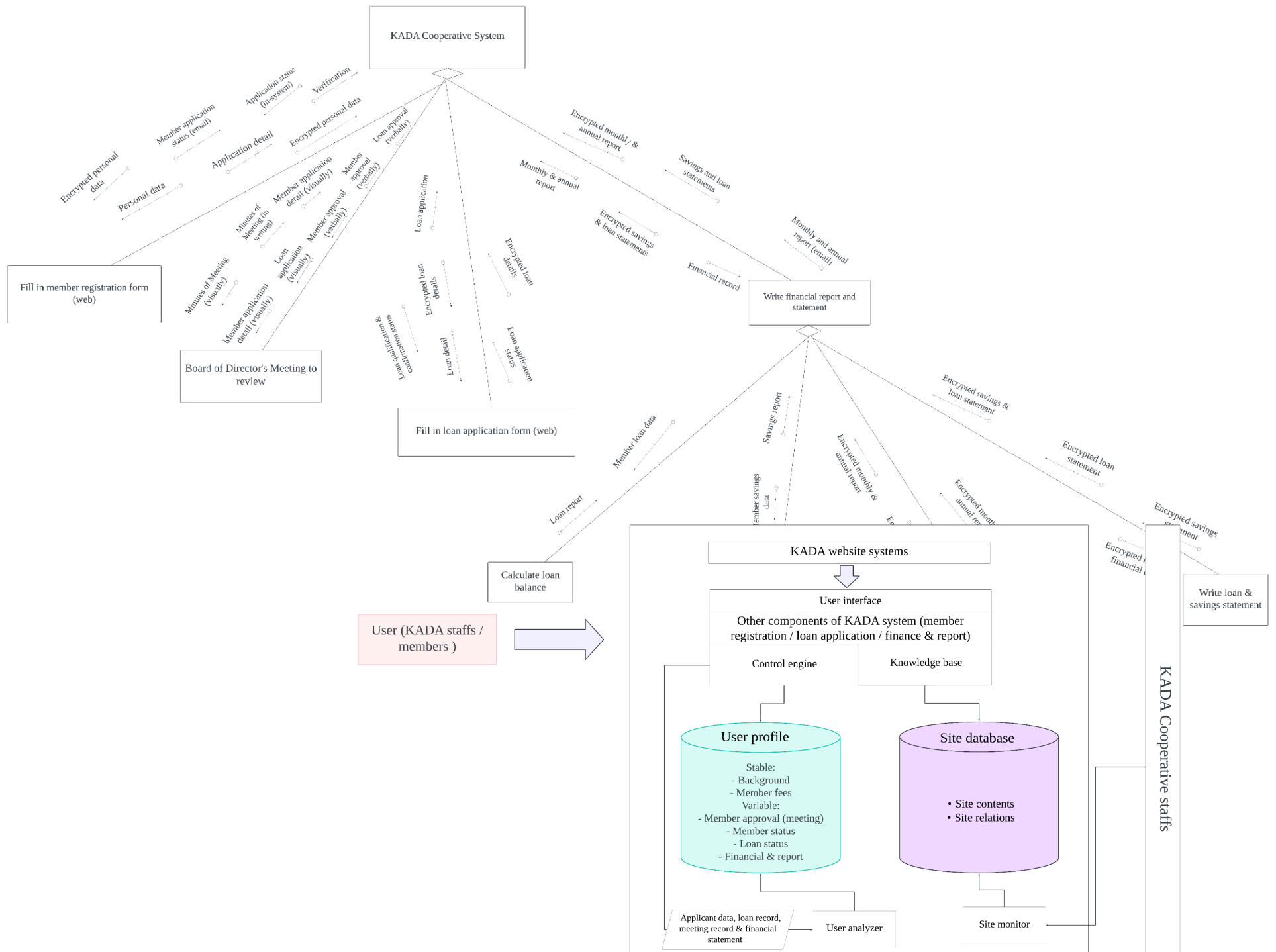
KADA Cooperative staffs

User (KADA staffs / members)

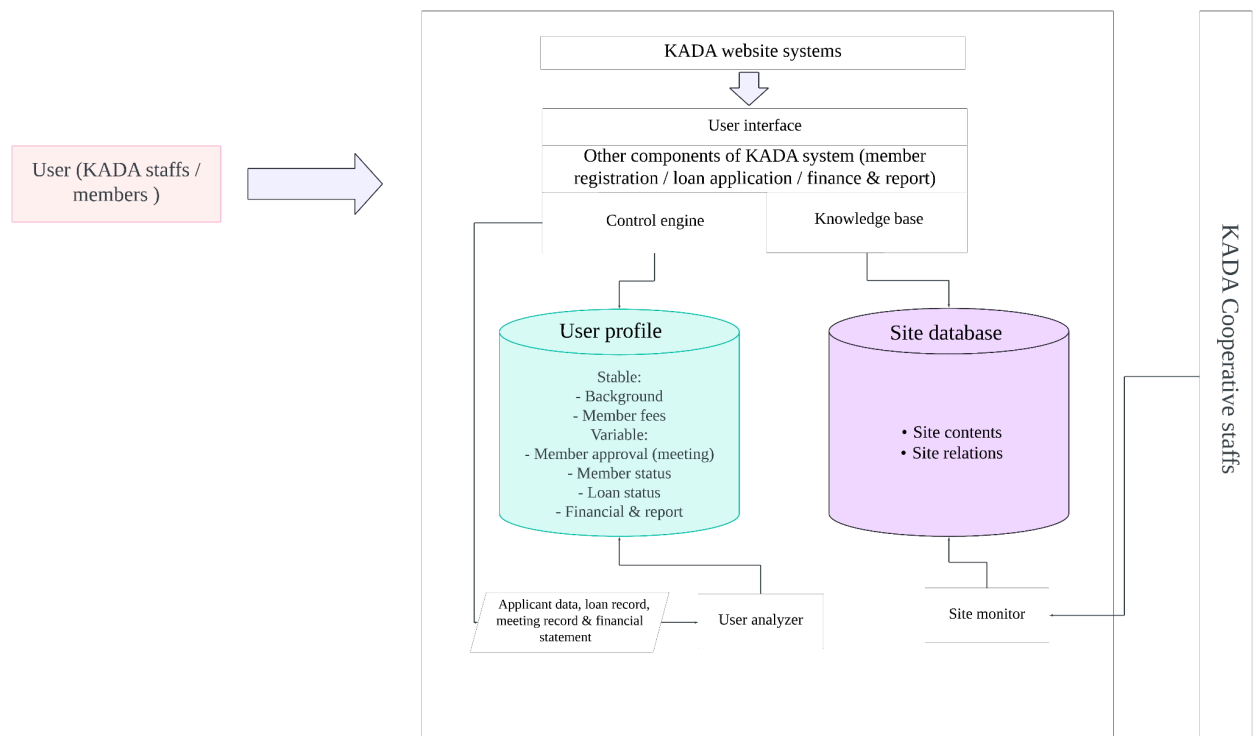


3.6 Structure Chart





3.7 System Architecture



4.0 System Wireframe

The wireframe

<https://www.figma.com/design/fmOM73PkW43IRRtUIv2Ky9/KADA-SYSTEM?node-id=0-1&t=X55UV1pICo5lsTLx-1>

The Flow of System Prototype for user, administrator and board of director

<https://www.figma.com/proto/fmOM73PkW43IRRtUIv2Ky9/SAD---KADA-SYSTEM?node-id=13-266&t=yKmZWa72YqFNJgYv-1&scaling=contain&content-scaling=fixed&page-id=0%3A1&starting-point-node-id=13%3A266>

